

Using Logarithmic Graphs

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Exponentials and Logarithms (ref e16)

This document contains all 90 *Using Logarithmic Graphs* questions, each with a fully worked solution and a TikZ graph. For the log-linear methods the diagram shows the *straight line in the transformed axes* ($\log y$ vs $\log x$, or $\log y/\ln y$ vs x) with the gradient, intercept and any data points marked, and crossovers for the comparison questions. For the sketching questions the actual curve is drawn with its intercepts and asymptotes labelled.

Contents

Using Logarithmic Graphs 01 (e16-001)	5
Using Logarithmic Graphs 02 (e16-002)	6
Using Logarithmic Graphs 03 (e16-003)	6
Using Logarithmic Graphs 04 (e16-004)	7
Using Logarithmic Graphs 05 (e16-005)	8
Using Logarithmic Graphs 06 (e16-006)	9
Using Logarithmic Graphs 07 (e16-007)	10
Using Logarithmic Graphs 08 (e16-008)	11
Using Logarithmic Graphs 09 (e16-009)	12
Using Logarithmic Graphs 10 (e16-010)	12
Using Logarithmic Graphs 11 (e16-011)	13
Using Logarithmic Graphs 12 (e16-012)	14
Using Logarithmic Graphs 13 (e16-013)	15
Using Logarithmic Graphs 14 (e16-014)	16
Using Logarithmic Graphs 15 (e16-015)	17

Using Logarithmic Graphs 16 (el6-016)	18
Using Logarithmic Graphs 17 (el6-017)	19
Using Logarithmic Graphs 18 (el6-018)	20
Using Logarithmic Graphs 19 (el6-019)	21
Using Logarithmic Graphs 20 (el6-020)	22
Using Logarithmic Graphs 21 (el6-021)	23
Using Logarithmic Graphs 22 (el6-022)	24
Using Logarithmic Graphs 23 (el6-023)	25
Using Logarithmic Graphs 24 (el6-024)	26
Using Logarithmic Graphs 25 (el6-025)	27
Using Logarithmic Graphs 26 (el6-026)	28
Using Logarithmic Graphs 27 (el6-027)	29
Using Logarithmic Graphs 28 (el6-028)	30
Using Logarithmic Graphs 29 (el6-029)	31
Using Logarithmic Graphs 30 (el6-030)	32
Using Logarithmic Graphs 31 (el6-031)	33
Using Logarithmic Graphs 32 (el6-032)	34
Using Logarithmic Graphs 33 (el6-033)	35
Using Logarithmic Graphs 34 (el6-034)	36
Using Logarithmic Graphs 35 (el6-035)	37
Using Logarithmic Graphs 36 (el6-036)	39
Using Logarithmic Graphs 37 (el6-037)	39
Using Logarithmic Graphs 38 (el6-038)	40
Using Logarithmic Graphs 39 (el6-039)	41

Using Logarithmic Graphs 40 (el6-040)	42
Using Logarithmic Graphs 41 (el6-041)	42
Using Logarithmic Graphs 42 (el6-042)	43
Using Logarithmic Graphs 43 (el6-043)	44
Using Logarithmic Graphs 44 (el6-044)	45
Using Logarithmic Graphs 45 (el6-045)	46
Using Logarithmic Graphs 46 (el6-046)	47
Using Logarithmic Graphs 47 (el6-047)	47
Using Logarithmic Graphs 48 (el6-048)	48
Using Logarithmic Graphs 49 (el6-049)	49
Using Logarithmic Graphs 50 (el6-050)	50
Using Logarithmic Graphs 51 (el6-051)	50
Using Logarithmic Graphs 52 (el6-052)	51
Using Logarithmic Graphs 53 (el6-053)	52
Using Logarithmic Graphs 54 (el6-054)	52
Using Logarithmic Graphs 55 (el6-055)	53
Using Logarithmic Graphs 56 (el6-056)	54
Using Logarithmic Graphs 57 (el6-057)	55
Using Logarithmic Graphs 58 (el6-058)	55
Using Logarithmic Graphs 59 (el6-059)	56
Using Logarithmic Graphs 60 (el6-060)	57
Using Logarithmic Graphs 61 (el6-061)	58
Using Logarithmic Graphs 62 (el6-062)	58
Using Logarithmic Graphs 63 (el6-063)	59

Using Logarithmic Graphs 64 (el6-064)	60
Using Logarithmic Graphs 65 (el6-065)	60
Using Logarithmic Graphs 66 (el6-066)	61
Using Logarithmic Graphs 67 (el6-067)	62
Using Logarithmic Graphs 68 (el6-068)	63
Using Logarithmic Graphs 69 (el6-069)	63
Using Logarithmic Graphs 70 (el6-070)	64
Using Logarithmic Graphs 71 (el6-071)	65
Using Logarithmic Graphs 72 (el6-072)	66
Using Logarithmic Graphs 73 (el6-073)	67
Using Logarithmic Graphs 74 (el6-074)	68
Using Logarithmic Graphs 75 (el6-075)	69
Using Logarithmic Graphs 76 (el6-076)	70
Using Logarithmic Graphs 77 (el6-077)	70
Using Logarithmic Graphs 78 (el6-078)	71
Using Logarithmic Graphs 79 (el6-079)	72
Using Logarithmic Graphs 80 (el6-080)	73
Using Logarithmic Graphs 81 (el6-081)	74
Using Logarithmic Graphs 82 (el6-082)	74
Using Logarithmic Graphs 83 (el6-083)	75
Using Logarithmic Graphs 84 (el6-084)	76
Using Logarithmic Graphs 85 (el6-085)	77
Using Logarithmic Graphs 86 (el6-086)	78
Using Logarithmic Graphs 87 (el6-087)	79

Using Logarithmic Graphs 01 (e16-001)

Difficulty: Foundation / Marks: 3

A relationship between x and y is believed to follow $y = ax^n$.

Show that taking $\log_{10}$ of both sides gives a straight-line equation of the form $\log y = n \log x + \log a$, and state what the gradient and vertical intercept represent.

Worked solution.

Step 1: Take $\log_{10}$ of both sides of $y = ax^n$.

$$\log y = \log(ax^n)$$

Applying the same log operation to both sides.

Step 2: Use the product law and power law.

$$\log y = \log a + \log(x^n) = \log a + n \log x$$

$$\log(ax^n) = \log a + \log(x^n) = \log a + n \log x.$$

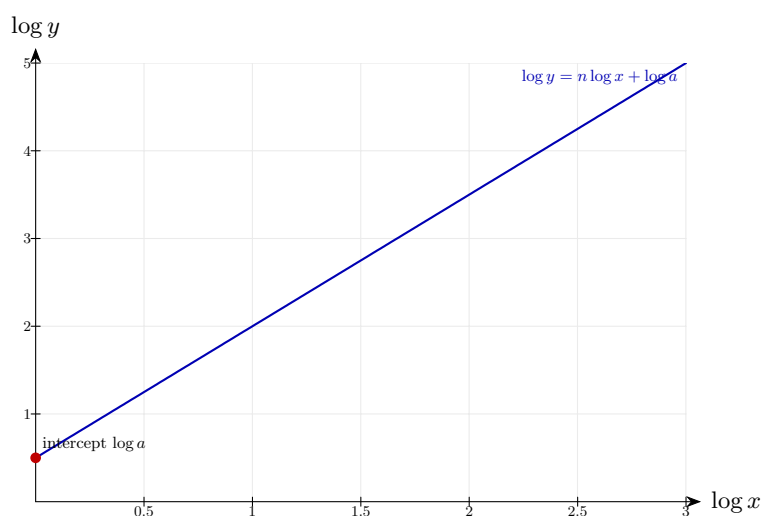
Step 3: Write in $Y = mX + c$ straight-line form.

$$\underbrace{\log y}_Y = \underbrace{n}_m \underbrace{\log x}_X + \underbrace{\log a}_c$$

Plotting $\log y$ against $\log x$ gives a straight line with gradient n and vertical intercept $\log a$.

Final answer: $\log y = n \log x + \log a$. Gradient = n ; vertical intercept = $\log a$.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 02 (e16-002)

Difficulty: Foundation / Marks: 3

A straight-line graph is obtained by plotting $\log_{10} y$ against $\log_{10} x$. The gradient is 3 and the vertical intercept is 0.5.

Find the values of a and n in the relationship $y = ax^n$.

Worked solution.

Step 1: The gradient of $\log y$ against $\log x$ equals n .

$$n = 3$$

For $y = ax^n$, the linearised form is $\log y = n \log x + \log a$, so gradient = n .

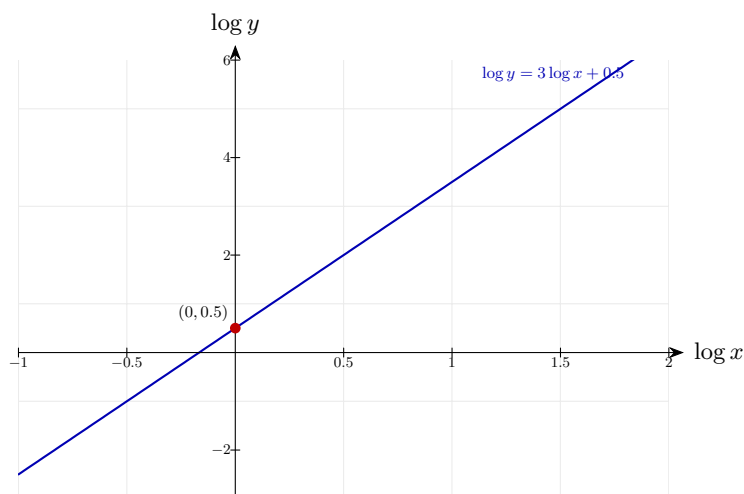
Step 2: The vertical intercept equals $\log a$.

$$\log a = 0.5 \Rightarrow a = 10^{0.5} = \sqrt{10} \approx 3.16$$

Apply $10^{\square}$ to find a .

Final answer: $n = 3, a = 10^{0.5} \approx 3.16$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 03 (e16-003)

Difficulty: Foundation / Marks: 4

A graph of $\log_{10} y$ against $\log_{10} x$ passes through the points (0, 1.2) and (2, 3.2).

Find the values of a and n in the equation $y = ax^n$.

Worked solution.

Step 1: Find the gradient of the straight line.

$$n = \frac{3.2 - 1.2}{2 - 0} = \frac{2.0}{2} = 1$$

$$\text{Gradient} = \Delta(\log y) / \Delta(\log x) = n.$$

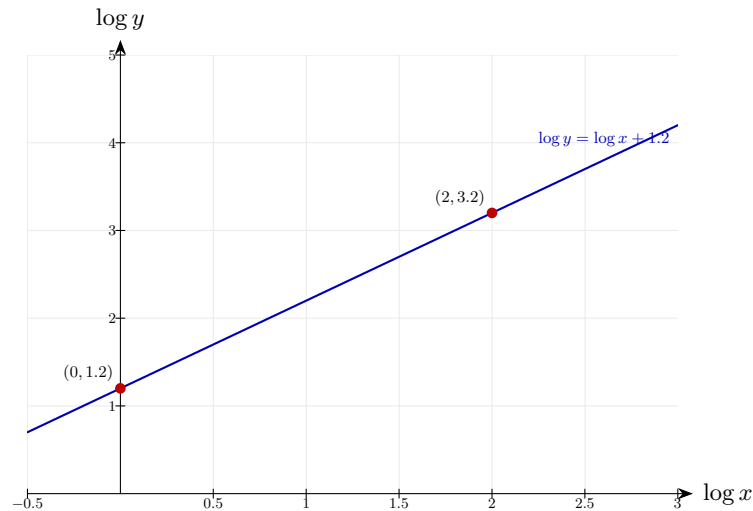
Step 2: The vertical intercept (where $\log x = 0$) gives $\log a$.

$$\log a = 1.2 \Rightarrow a = 10^{1.2} \approx 15.8$$

The point $(0, 1.2)$ is on the $\log y$ -axis, so this is the intercept.

Final answer: $n = 1, a \approx 15.8$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 04 (e16-004)

Difficulty: Foundation / Marks: 4

A graph of $\log_{10} y$ against $\log_{10} x$ passes through $(1, 2.6)$ and $(3, 4.0)$.

Find a and n in $y = ax^n$. Give your answers to 3 s.f.

Worked solution.

Step 1: Find the gradient n .

$$n = \frac{4.0 - 2.6}{3 - 1} = \frac{1.4}{2} = 0.7$$

Gradient = change in $\log y$ / change in $\log x$.

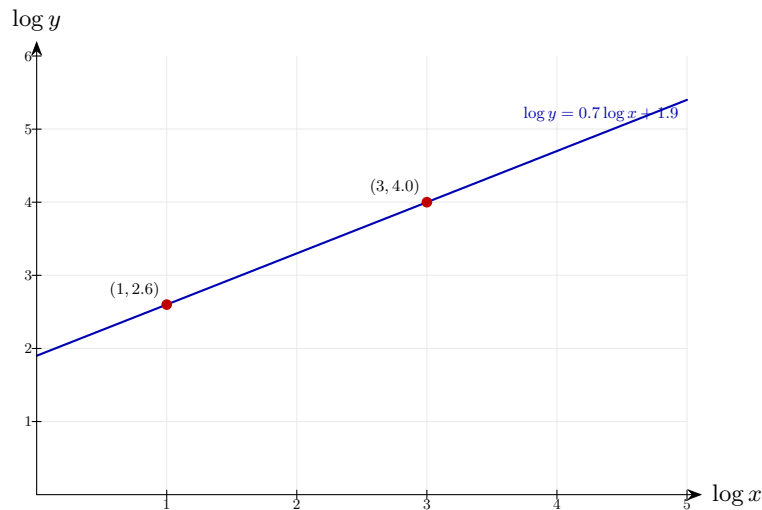
Step 2: Find the vertical intercept using point $(1, 2.6)$.

$$\log a = 2.6 - 0.7 \times 1 = 1.9 \Rightarrow a = 10^{1.9} \approx 79.4$$

Rearrange $\log y = n \log x + \log a$ using one of the points.

Final answer: $n = 0.7, a \approx 79.4$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 05 (e16-005)

Difficulty: Foundation / Marks: 4

The table below shows values of x and y that are thought to satisfy $y = ax^n$.

x	4	9	16	25
y	24	81	192	375

By plotting $\log y$ against $\log x$, find a and n .

Worked solution.

Step 1: Convert the data to $\log x$ and $\log y$. Using $x = 4, y = 24$ and $x = 25, y = 375$:

$$\log 4 \approx 0.602, \log 24 \approx 1.380 \quad \log 25 \approx 1.398, \log 375 \approx 2.574$$

Taking $\log_{10}$ of each data value.

Step 2: Calculate the gradient n .

$$n = \frac{2.574 - 1.380}{1.398 - 0.602} = \frac{1.194}{0.796} \approx 1.50$$

Gradient = change in $\log y$ / change in $\log x$.

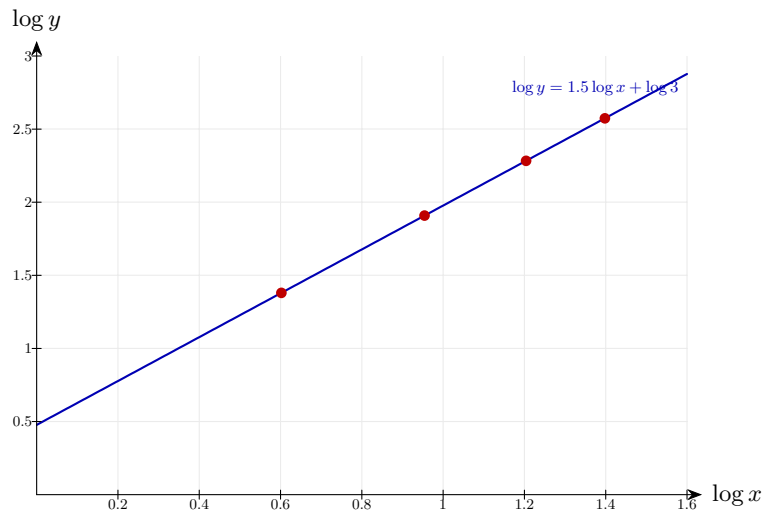
Step 3: Find $\log a$ using the intercept.

$$\log a = 1.380 - 1.50 \times 0.602 = 1.380 - 0.903 = 0.477 \Rightarrow a = 10^{0.477} \approx 3$$

Substitute one point into $\log y = n \log x + \log a$.

Final answer: $n \approx 1.5, a \approx 3$, so $y \approx 3x^{1.5}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 06 (e16-006)

Difficulty: Foundation / Marks: 3

The radius r cm and height h cm of several cylinders are thought to satisfy $h = ar^n$. A graph of $\log h$ against $\log r$ gives a straight line with gradient -2 and vertical intercept 3 . Write down the relationship between h and r .

Worked solution.

Step 1: Read off n from the gradient.

$$n = -2$$

Gradient of $\log h$ vs $\log r$ equals the power n .

Step 2: Find a from the vertical intercept.

$$\log a = 3 \Rightarrow a = 10^3 = 1000$$

The vertical intercept of the linearised graph equals $\log a$.

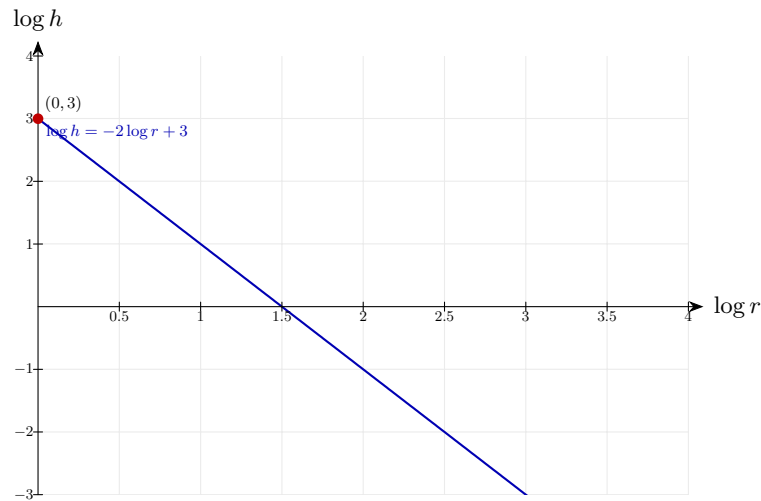
Step 3: Write the relationship.

$$h = 1000r^{-2} = \frac{1000}{r^2}$$

Substitute $a = 1000$ and $n = -2$.

Final answer: $h = \frac{1000}{r^2}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 07 (e16-007)

Difficulty: Foundation / Marks: 3

A relationship between x and y is believed to follow $y = kb^x$.

Show that taking $\log_{10}$ of both sides gives a straight-line equation of the form $\log y = x \log b + \log k$, and state what the gradient and vertical intercept represent.

Worked solution.

Step 1: Take $\log_{10}$ of both sides.

$$\log y = \log(kb^x)$$

Same log applied to both sides.

Step 2: Apply the product and power laws.

$$\log y = \log k + \log(b^x) = \log k + x \log b$$

$$\log(kb^x) = \log k + x \log b.$$

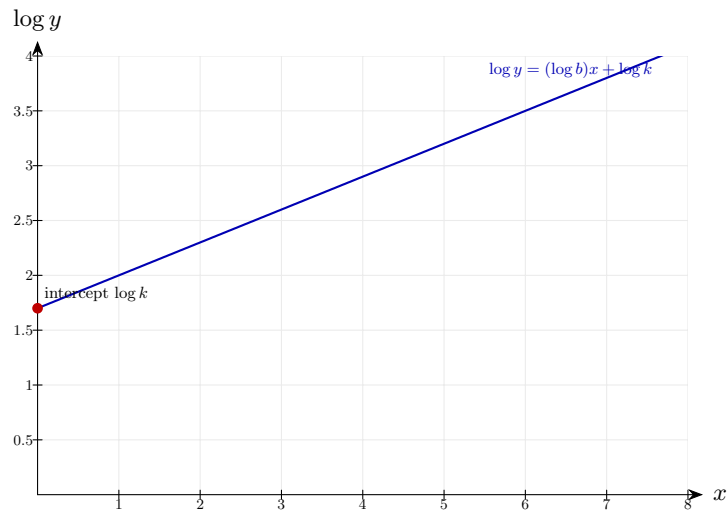
Step 3: Identify the straight-line form.

$$\underbrace{\log y}_Y = \underbrace{\log b}_m \underbrace{x}_X + \underbrace{\log k}_c$$

Plotting $\log y$ against x gives a straight line with gradient $\log b$ and vertical intercept $\log k$.

Final answer: $\log y = x \log b + \log k$. Gradient = $\log b$; vertical intercept = $\log k$.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 08 (e16-008)

Difficulty: Foundation / Marks: 3

A graph of $\log_{10} y$ against x is a straight line with gradient 0.3 and vertical intercept 1.7. Find the values of k and b in the equation $y = kb^x$. Give your answers to 3 s.f.

Worked solution.

Step 1: The gradient equals $\log b$.

$$\log b = 0.3 \Rightarrow b = 10^{0.3} \approx 2.00$$

Apply $10^{\square}$ to find b .

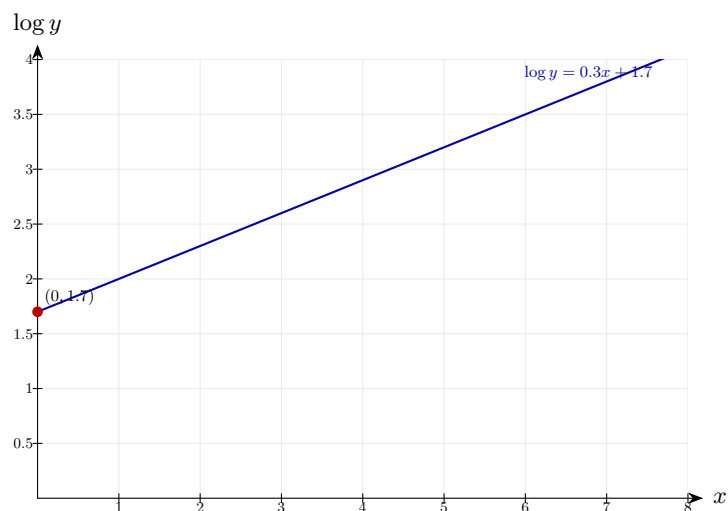
Step 2: The vertical intercept equals $\log k$.

$$\log k = 1.7 \Rightarrow k = 10^{1.7} \approx 50.1$$

Apply $10^{\square}$ to find k .

Final answer: $b \approx 2.00$, $k \approx 50.1$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 09 (e16-009)

Difficulty: Foundation / Marks: 4

A graph of $\log_{10} y$ against x passes through the points $(0, 2)$ and $(5, 3.5)$.

Find k and b in $y = kb^x$. Give your answers to 3 s.f.

Worked solution.

Step 1: Find the gradient $\log b$.

$$\log b = \frac{3.5 - 2}{5 - 0} = \frac{1.5}{5} = 0.3 \Rightarrow b = 10^{0.3} \approx 2.00$$

Gradient = change in $\log y$ / change in x .

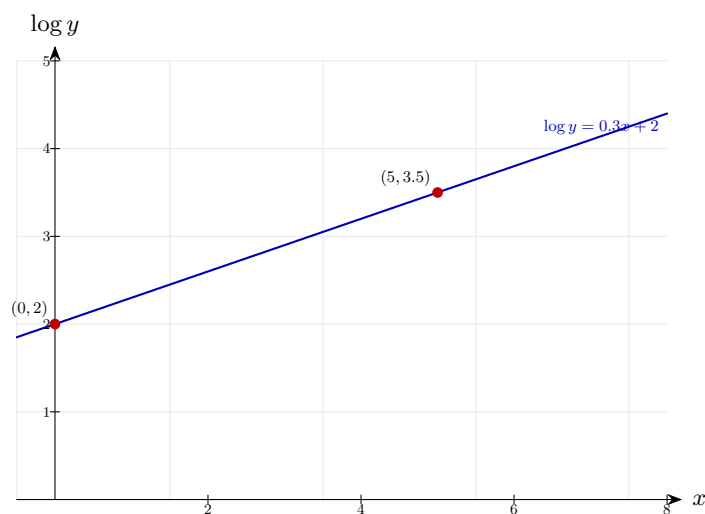
Step 2: The vertical intercept gives $\log k$.

$$\log k = 2 \Rightarrow k = 10^2 = 100$$

The point $(0, 2)$ is on the $\log y$ -axis, so it is the intercept.

Final answer: $b \approx 2.00$, $k = 100$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 10 (e16-010)

Difficulty: Foundation / Marks: 4

A graph of $\log_{10} y$ against x passes through $(2, 1.4)$ and $(6, 2.8)$.

Find k and b in $y = kb^x$. Give your answers to 3 s.f.

Worked solution.

Step 1: Find the gradient $\log b$.

$$\log b = \frac{2.8 - 1.4}{6 - 2} = \frac{1.4}{4} = 0.35 \Rightarrow b = 10^{0.35} \approx 2.24$$

Gradient = $\Delta(\log y)/\Delta x$.

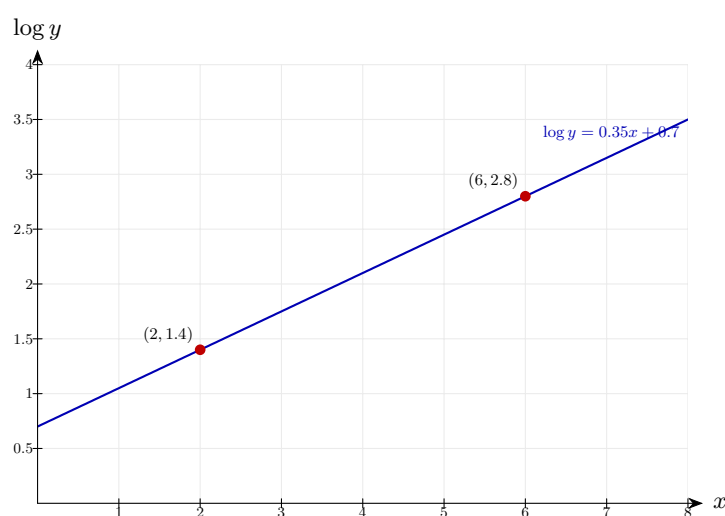
Step 2: Find $\log k$ using point $(2, 1.4)$.

$$1.4 = 0.35 \times 2 + \log k \Rightarrow \log k = 1.4 - 0.7 = 0.7 \Rightarrow k = 10^{0.7} \approx 5.01$$

Substitute into $\log y = x \log b + \log k$.

Final answer: $b \approx 2.24$, $k \approx 5.01$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 11 (e16-011)

Difficulty: Foundation / Marks: 3

The equation $y = kb^x$ can also be written as $y = ke^{cx}$ for some constant c .

A graph of $\ln y$ against x has gradient 0.4 and vertical intercept 3.

Write down the values of k and c in $y = ke^{cx}$.

Worked solution.

Step 1: Linearise $y = ke^{cx}$ using $\ln$.

$$\ln y = cx + \ln k$$

$$\ln(ke^{cx}) = \ln k + cx.$$

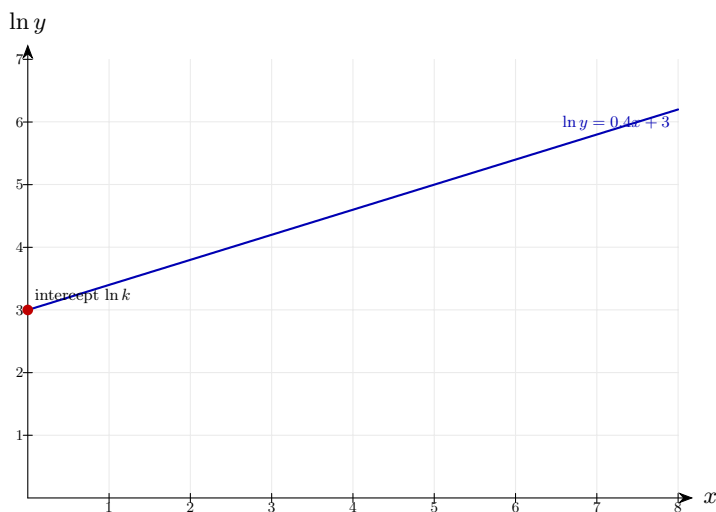
Step 2: Read off the gradient c and intercept $\ln k$.

$$c = 0.4 \quad \ln k = 3 \Rightarrow k = e^3 \approx 20.1$$

Gradient = c ; intercept = $\ln k$.

Final answer: $c = 0.4$, $k = e^3 \approx 20.1$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 12 (e16-012)

Difficulty: Foundation / Marks: 4

A graph of $\ln y$ against x passes through $(0, 1.5)$ and $(4, 3.9)$.

Find k and c in $y = ke^{cx}$. Give your answers to 3 s.f.

Worked solution.

Step 1: Find the gradient c .

$$c = \frac{3.9 - 1.5}{4 - 0} = \frac{2.4}{4} = 0.6$$

Gradient of $\ln y$ against x .

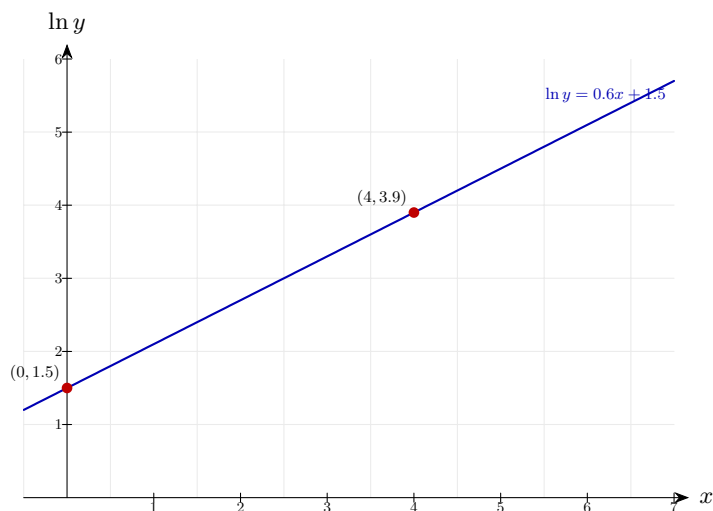
Step 2: The vertical intercept is $\ln k$.

$$\ln k = 1.5 \Rightarrow k = e^{1.5} \approx 4.48$$

Point $(0, 1.5)$ is on the $\ln y$ -axis.

Final answer: $c = 0.6$, $k \approx 4.48$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 13 (e16-013)

Difficulty: Foundation / Marks: 5

The table below gives values of t and y which satisfy $y = kb^t$.

t	1	3	5	7
y	6	30	150	750

- Copy and complete the table of $\log_{10} y$ values.
- State the gradient and vertical intercept of the graph of $\log y$ against t .
- Hence find k and b .

Worked solution.

Step 1: (a) Compute $\log_{10} y$ for each value.

$$\log 6 \approx 0.778, \quad \log 30 \approx 1.477, \quad \log 150 \approx 2.176, \quad \log 750 \approx 2.875$$

Use a calculator to find each log value to 3 d.p.

Step 2: (b) Find the gradient using two points, e.g. (1, 0.778) and (7, 2.875).

$$\log b = \frac{2.875 - 0.778}{7 - 1} = \frac{2.097}{6} \approx 0.350$$

Gradient = $\log b$.

Step 3: Find the vertical intercept (where $t = 0$).

$$\log k = 0.778 - 0.350 \times 1 = 0.428$$

Substitute into $\log y = t \log b + \log k$.

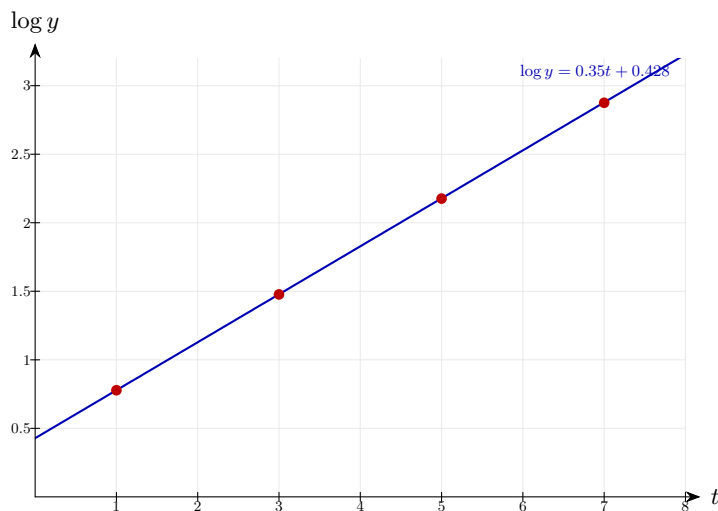
Step 4: (c) Find k and b .

$$b = 10^{0.350} \approx 2.24 \quad k = 10^{0.428} \approx 2.68$$

Apply $10^{\square}$ to each log value.

Final answer: (b) Gradient ≈ 0.350 , intercept ≈ 0.428 (c) $k \approx 2.68$, $b \approx 2.24$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 14 (e16-014)

Difficulty: Foundation / Marks: 5

The table below gives values of x and y which satisfy $y = ax^n$.

x	2	4	8	16
y	12	34	96	271

- Complete the table of $\log x$ and $\log y$ values.
- Find the gradient and vertical intercept of $\log y$ against $\log x$.
- Find a and n to 3 s.f.

Worked solution.

Step 1: (a) Compute $\log x$ and $\log y$.

$\log x$	0.301	0.602	0.903	1.204
$\log y$	1.079	1.531	1.982	2.433

Take $\log_{10}$ of each value.

Step 2: (b) Calculate the gradient using the first and last points.

$$n = \frac{2.433 - 1.079}{1.204 - 0.301} = \frac{1.354}{0.903} \approx 1.50$$

Gradient of $\log y$ vs $\log x = n$.

Step 3: Find the intercept.

$$\log a = 1.079 - 1.50 \times 0.301 = 1.079 - 0.452 = 0.627$$

Substitute the first point.

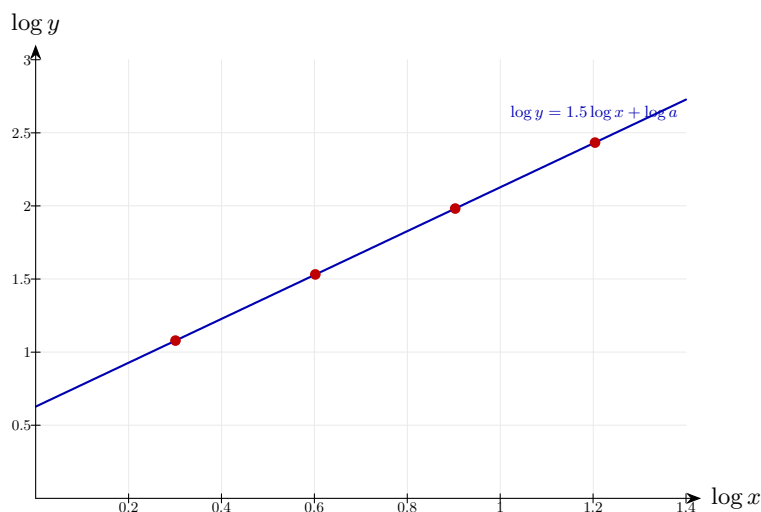
Step 4: (c) Find a and n .

$$n \approx 1.50, \quad a = 10^{0.627} \approx 4.24$$

$$10^{0.627} \approx 4.24.$$

Final answer: (c) $n \approx 1.50$, $a \approx 4.24$, so $y \approx 4.24x^{1.5}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 15 (e16-015)

Difficulty: Foundation / Marks: 5

The activity x (Bq) of a radioactive substance at time t (days) satisfies $x = kb^t$. The measurements are:

t	10	20	40	80
x	500	250	62.5	15.6

- Find $\log_{10} x$ for each value of t .
- Plot $\log x$ against t and draw a line of best fit.
- Find k and b . Give your answers to 3 s.f.

Worked solution.

Step 1: (a) Compute $\log x$ for each value.

$$\log 500 \approx 2.699, \quad \log 250 \approx 2.398, \quad \log 62.5 \approx 1.796, \quad \log 15.6 \approx 1.193$$

Use a calculator.

Step 2: (c) Gradient using points (10, 2.699) and (80, 1.193).

$$\log b = \frac{1.193 - 2.699}{80 - 10} = \frac{-1.506}{70} \approx -0.02151$$

Gradient = $\log b$; negative as this is decay.

Step 3: Find $\log k$ from the intercept.

$$\log k = 2.699 - (-0.02151)(10) = 2.699 + 0.2151 = 2.914 \Rightarrow k = 10^{2.914} \approx 821$$

Substitute the first point into $\log x = t \log b + \log k$.

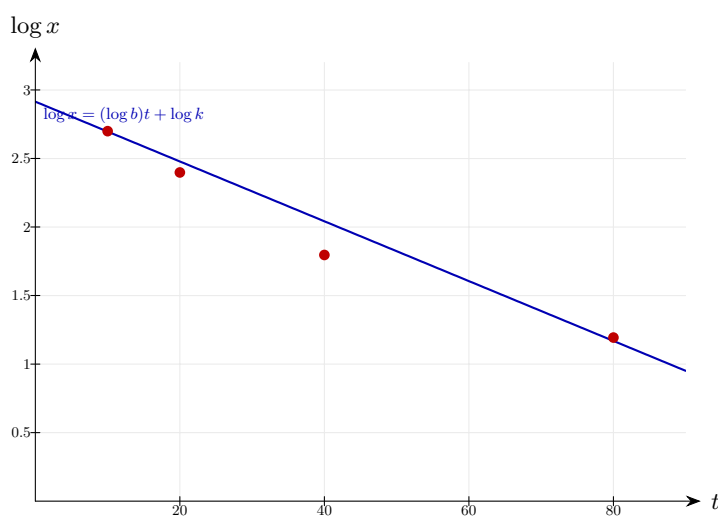
Step 4: Find b .

$$b = 10^{-0.02151} \approx 0.951$$

$b < 1$ confirms this is decay.

Final answer: (c) $k \approx 821$, $b \approx 0.951$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 16 (e16-016)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against $\log x$ gives a straight line with equation $\log y = 2 \log x + 0.6$.

(a) Find y when $x = 5$.

(b) Find x when $y = 200$. Give your answers to 3 s.f.

Worked solution.

Step 1: First identify a and n .

$$n = 2, \quad a = 10^{0.6} \approx 3.98, \quad \text{so } y \approx 3.98x^2$$

Reading gradient and intercept from the line equation.

Step 2: (a) Substitute $x = 5$.

$$y \approx 3.98 \times 25 = 99.5$$

Or substitute into the log equation: $\log y = 2 \log 5 + 0.6 = 2(0.699) + 0.6 = 1.998 \Rightarrow y = 10^{1.998} \approx 99.5$.

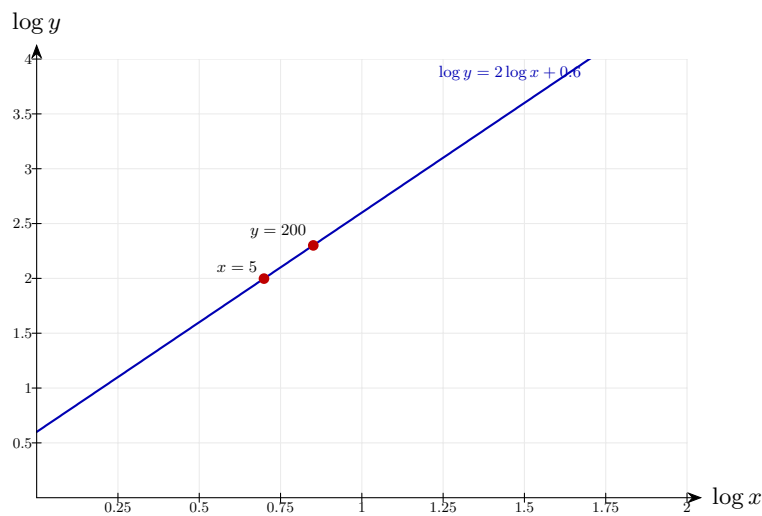
Step 3: (b) Set $y = 200$ and solve.

$$\log 200 = 2 \log x + 0.6 \Rightarrow 2.301 = 2 \log x + 0.6 \Rightarrow \log x = \frac{1.701}{2} = 0.8505 \Rightarrow x = 10^{0.8505} \approx 7.09$$

Substitute $\log y = \log 200$ and solve for $\log x$.

Final answer: (a) $y \approx 99.5$ (b) $x \approx 7.09$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 17 (e16-017)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against x gives the equation $\log y = 0.2x + 1.3$.

(a) Find y when $x = 3$.

(b) Find x when $y = 1000$. Give your answers to 3 s.f.

Worked solution.

Step 1: (a) Substitute $x = 3$.

$$\log y = 0.2(3) + 1.3 = 0.6 + 1.3 = 1.9 \Rightarrow y = 10^{1.9} \approx 79.4$$

Direct substitution into the linearised equation.

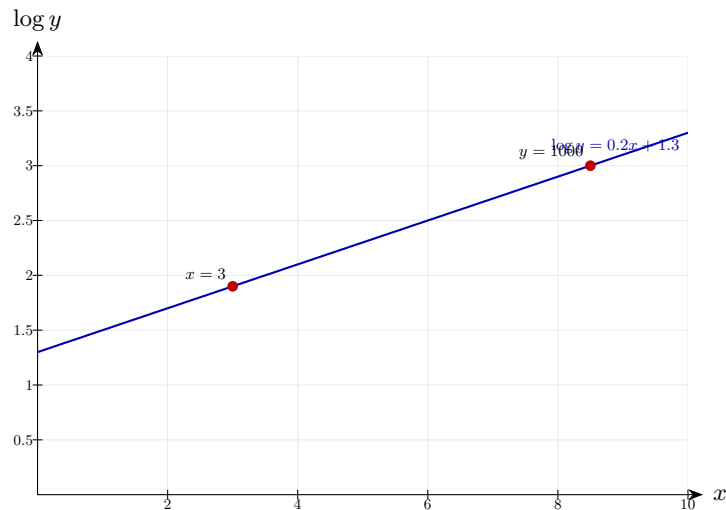
Step 2: (b) Set $y = 1000$.

$$\log(1000) = 0.2x + 1.3 \Rightarrow 3 = 0.2x + 1.3 \Rightarrow 0.2x = 1.7 \Rightarrow x = 8.5$$

$\log 1000 = 3$. Solve the resulting linear equation.

Final answer: (a) $y \approx 79.4$ (b) $x = 8.5$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 18 (e16-018)

Difficulty: Foundation / Marks: 5

The number of subscribers S to a blog after t weeks satisfies $S = at^b$. A graph of $\log S$ against $\log t$ is a straight line passing through $(0.5, 2.1)$ and $(1.5, 3.3)$.

(a) Find a and b .

(b) Predict the number of subscribers after 20 weeks.

Worked solution.

Step 1: (a) Find the gradient b .

$$b = \frac{3.3 - 2.1}{1.5 - 0.5} = \frac{1.2}{1.0} = 1.2$$

Gradient = $n = b$ in this model.

Step 2: Find $\log a$ using point $(0.5, 2.1)$.

$$2.1 = 1.2 \times 0.5 + \log a \Rightarrow \log a = 2.1 - 0.6 = 1.5 \Rightarrow a = 10^{1.5} \approx 31.6$$

Substitute into $\log S = b \log t + \log a$.

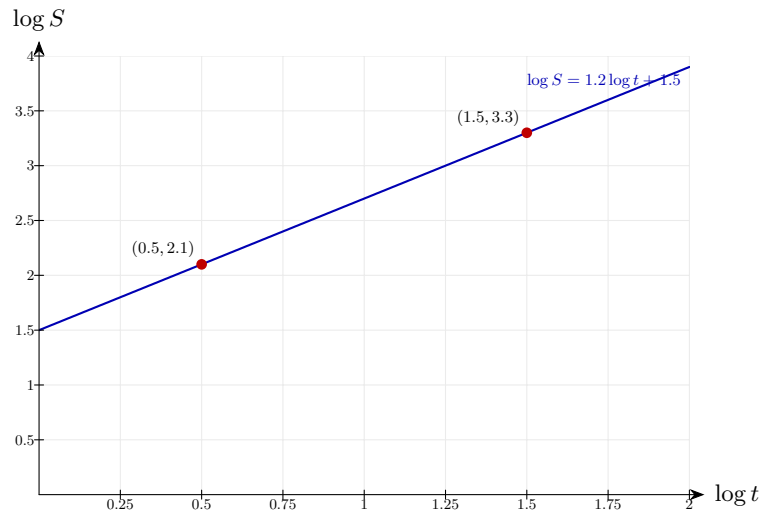
Step 3: (b) Predict at $t = 20$.

$$S = 31.6 \times 20^{1.2} \approx 31.6 \times 36.41 \approx 1150$$

$20^{1.2} \approx 36.41$; evaluate using a calculator. Equivalently, $\log S = 1.2 \log 20 + 1.5 = 1.2(1.301) + 1.5 = 3.061 \Rightarrow S \approx 10^{3.061} \approx 1150$.

Final answer: (a) $b = 1.2$, $a \approx 31.6$ (b) ≈ 1150 subscribers

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 19 (e16-019)

Difficulty: Foundation / Marks: 5

The value £ V of a machine t years after purchase satisfies $V = pq^t$. A graph of $\log V$ against t passes through $(0, 4.3)$ and $(10, 3.1)$.

(a) Find p and q to 3 s.f.

(b) Find the value of the machine after 7 years.

Worked solution.

Step 1: (a) Vertical intercept gives $\log p$.

$$\log p = 4.3 \Rightarrow p = 10^{4.3} \approx 19\,953$$

Point $(0, 4.3)$ is on the $\log V$ -axis.

Step 2: Gradient gives $\log q$.

$$\log q = \frac{3.1 - 4.3}{10 - 0} = \frac{-1.2}{10} = -0.12 \Rightarrow q = 10^{-0.12} \approx 0.759$$

Gradient = $\log q$; negative since this is depreciation.

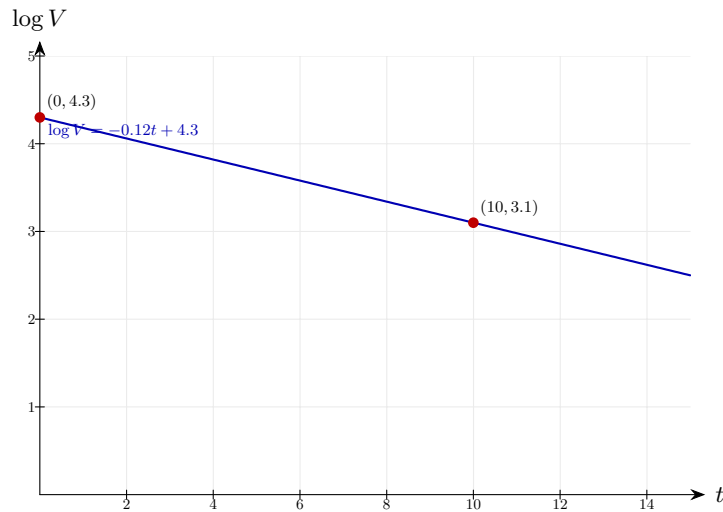
Step 3: (b) Substitute $t = 7$.

$$\log V = -0.12(7) + 4.3 = -0.84 + 4.3 = 3.46 \Rightarrow V = 10^{3.46} \approx \text{£}2884$$

Use the linearised equation directly: substitute $t = 7$ and apply $10^{\square}$. Equivalently, $V = 19\,953 \times 0.759^7 \approx 19\,953 \times 0.145 \approx \text{£}2890$ (small difference due to rounding q).

Final answer: (a) $p \approx 19\,953$, $q \approx 0.759$ (b) $\approx \text{£}2884$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 20 (e16-020)

Difficulty: Foundation / Marks: 5

The pressure P (Pa) and volume V (m^3) of a gas satisfy $P = kV^n$. A graph of $\log P$ against $\log V$ has gradient -1.4 and passes through the point $(0.5, 3.8)$.

(a) Find k and n .

(b) Find P when $V = 0.2$. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Gradient = n .

$$n = -1.4$$

Read directly from the gradient.

Step 2: Find $\log k$ using point $(0.5, 3.8)$.

$$3.8 = -1.4(0.5) + \log k \Rightarrow \log k = 3.8 + 0.7 = 4.5 \Rightarrow k = 10^{4.5} \approx 31\,623$$

Substitute into the linearised equation.

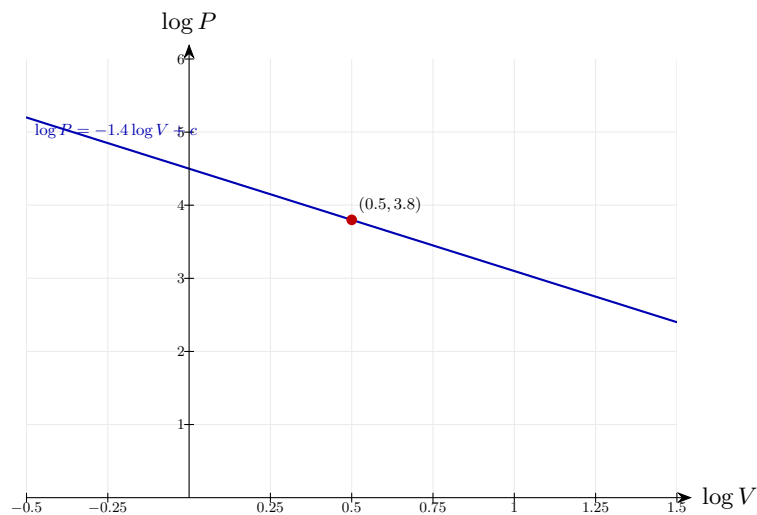
Step 3: (b) Substitute $V = 0.2$.

$$\log P = -1.4 \log(0.2) + 4.5 = -1.4(-0.699) + 4.5 = 0.979 + 4.5 = 5.479 \Rightarrow P = 10^{5.479} \approx 301\,000$$

$$\log 0.2 = \log(1/5) = -\log 5 \approx -0.699.$$

Final answer: (a) $n = -1.4$, $k \approx 31\,623$ (b) $P \approx 301\,000$ Pa

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 21 (e16-021)

Difficulty: Foundation / Marks: 3

The relationship between x and y is $y = 5 \times 4^x$.

- (a) Show that a graph of $\log y$ against x will be a straight line.
 (b) State the gradient and vertical intercept of this straight line.

Worked solution.

Step 1: (a) Take $\log_{10}$ of $y = 5 \times 4^x$.

$$\log y = \log 5 + x \log 4$$

Using the product law and power law.

Step 2: Identify the straight-line form.

$$\log y = (\log 4)x + \log 5$$

This is $Y = mX + c$ where $Y = \log y$ and $X = x$. Since m and c are constants, this is a straight line.
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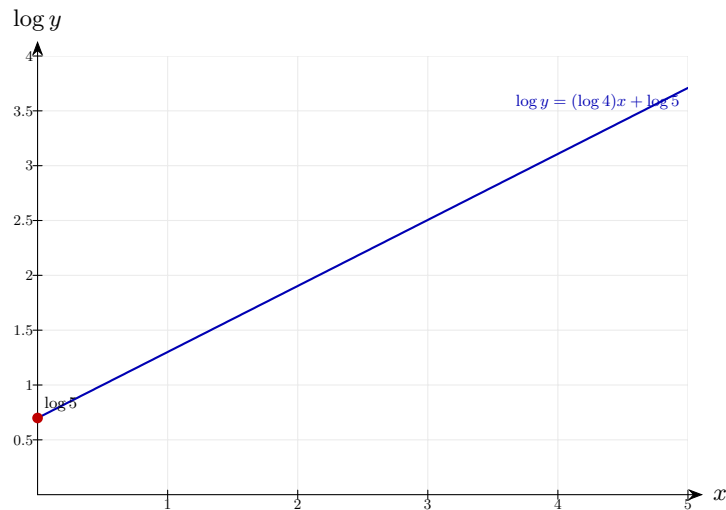
Step 3: (b) State gradient and intercept.

$$\text{Gradient} = \log 4 \approx 0.602 \quad \text{Intercept} = \log 5 \approx 0.699$$

Evaluate numerically.

Final answer: (b) Gradient = $\log 4 \approx 0.602$; vertical intercept = $\log 5 \approx 0.699$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 22 (e16-022)

Difficulty: Foundation / Marks: 4

The height h (m) and mass m (kg) of several trees satisfy $h = 2m^{0.4}$.

- Write the equation in linear form suitable for plotting $\log h$ against $\log m$.
- State the gradient and vertical intercept.
- Describe the graph of h against m .

Worked solution.

Step 1: (a) Take log of both sides.

$$\log h = 0.4 \log m + \log 2$$

Using the power law and product law.

Step 2: (b) State gradient and intercept.

$$\text{Gradient} = 0.4 \quad \text{Intercept} = \log 2 \approx 0.301$$

Read directly from the linearised equation.

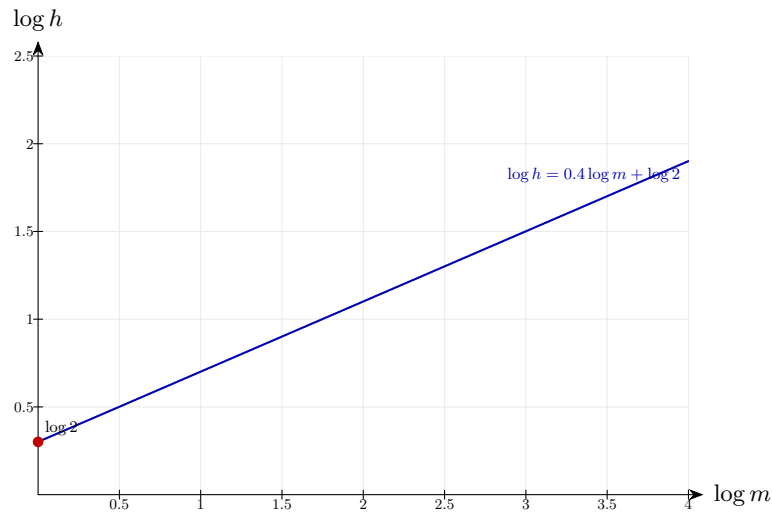
Step 3: (c) The original graph $h = 2m^{0.4}$ is a curve.

$$h = 2m^{0.4} \text{ - a power curve passing through } (1, 2)$$

Since $0 < n < 1$, the curve is concave (increasing but at a decreasing rate).

Final answer: (a) $\log h = 0.4 \log m + \log 2$ (b) Gradient 0.4; intercept ≈ 0.301 (c) An increasing concave power curve.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 23 (e16-023)

Difficulty: Foundation / Marks: 3

A tank is being filled with water. After s seconds the height h mm of water satisfies $h = 3 \times 2^s$.

(a) Write this in linear form for a graph of $\log h$ against s .

(b) Explain which graph — h against s or $\log h$ against s — is more useful for calculations.

Worked solution.

Step 1: (a) Take log of $h = 3 \times 2^s$.

$$\log h = s \log 2 + \log 3$$

$$\log(3 \times 2^s) = \log 3 + s \log 2.$$

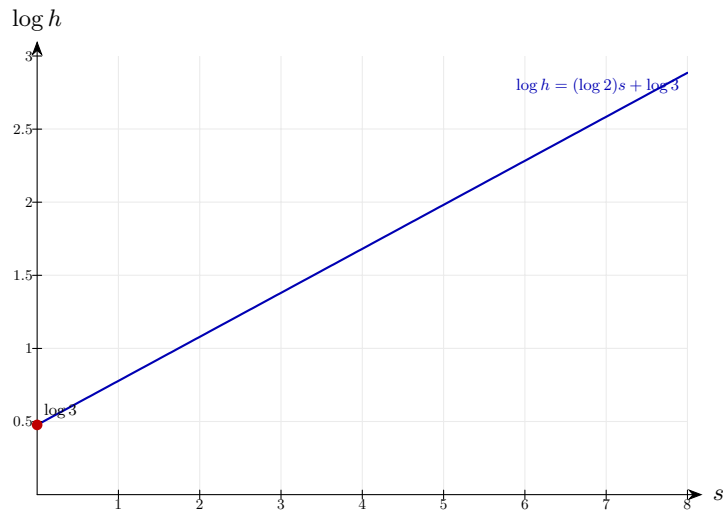
Step 2: (b) Compare the two graphs.

$\log h$ vs s is a straight line; h vs s is an exponential curve.

A straight line is easier to work with: gradient and intercept can be read off directly, and the line of best fit is more reliable. Reading values off a straight line is more accurate than off a curve.

Final answer: (a) $\log h = s \log 2 + \log 3$ (b) The graph of $\log h$ against s is more useful — it is a straight line, making values and the line of best fit easier to determine.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 24 (e16-024)

Difficulty: Foundation / Marks: 3

The population p of bacteria after t hours satisfies $p = 50e^{0.3t}$.

- Write this in linear form for a graph of $\ln p$ against t .
- State the gradient and vertical intercept of the straight-line graph.

Worked solution.

Step 1: (a) Take $\ln$ of both sides.

$$\ln p = \ln(50e^{0.3t}) = \ln 50 + 0.3t$$

Using the product law: $\ln(50e^{0.3t}) = \ln 50 + \ln(e^{0.3t}) = \ln 50 + 0.3t$.

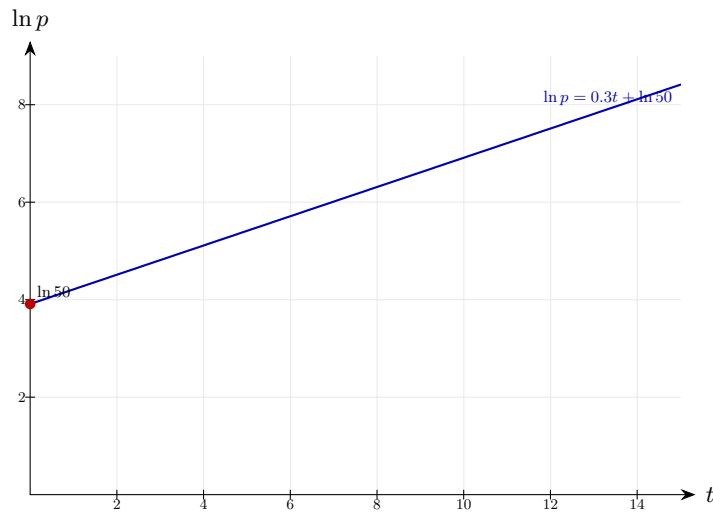
Step 2: (b) Identify gradient and intercept.

$$\text{Gradient} = 0.3 \quad \text{Intercept} = \ln 50 \approx 3.91$$

Gradient = coefficient of t ; intercept = $\ln k$.

Final answer: (a) $\ln p = 0.3t + \ln 50$ (b) Gradient = 0.3; intercept = $\ln 50 \approx 3.91$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 25 (e16-025)

Difficulty: Foundation / Marks: 5

Two populations are modelled by $\ln P$ against t graphs.

Population A: $\ln P = 0.2t + 4$

Population B: $\ln P = -0.1t + 7$

(a) Find the time t when the populations are equal.

(b) Find the common population at this time. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) The populations are equal when the lines intersect.

$$0.2t + 4 = -0.1t + 7$$

Set the two expressions for $\ln P$ equal.

Step 2: Solve for t .

$$0.3t = 3 \Rightarrow t = 10$$

Collect t terms and divide.

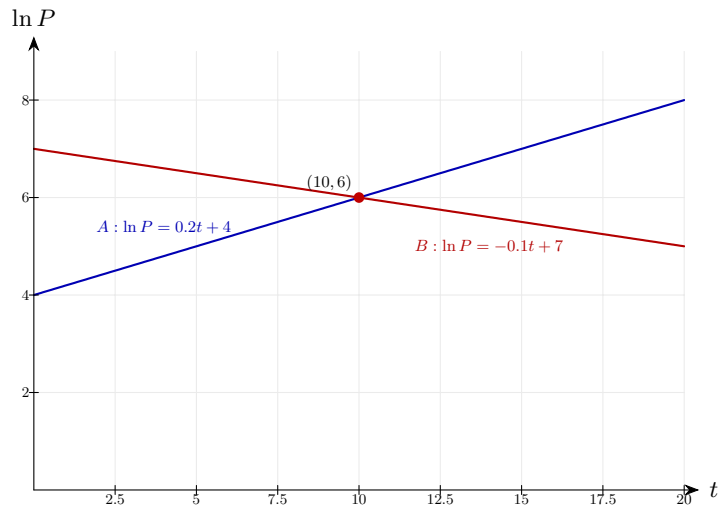
Step 3: (b) Find the common $\ln P$ and then P .

$$\ln P = 0.2(10) + 4 = 6 \Rightarrow P = e^6 \approx 403$$

Substitute $t = 10$ into either equation.

Final answer: (a) $t = 10$ (b) $P = e^6 \approx 403$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 26 (e16-026)

Difficulty: Foundation / Marks: 5

Two rival blogs have subscriber counts modelled as follows (where t is days):

Blog A (growing): $\log S = 0.08t + 2$

Blog B (declining): $\log S = -0.04t + 3.5$

(a) Find the number of subscribers each blog starts with.

(b) Find the day on which both blogs have the same number of subscribers.

Worked solution.

Step 1: (a) At $t = 0$, find each initial subscriber count.

$$\text{Blog A: } \log S = 2 \Rightarrow S = 100 \quad \text{Blog B: } \log S = 3.5 \Rightarrow S = 10^{3.5} \approx 3162$$

Substitute $t = 0$ into each equation.

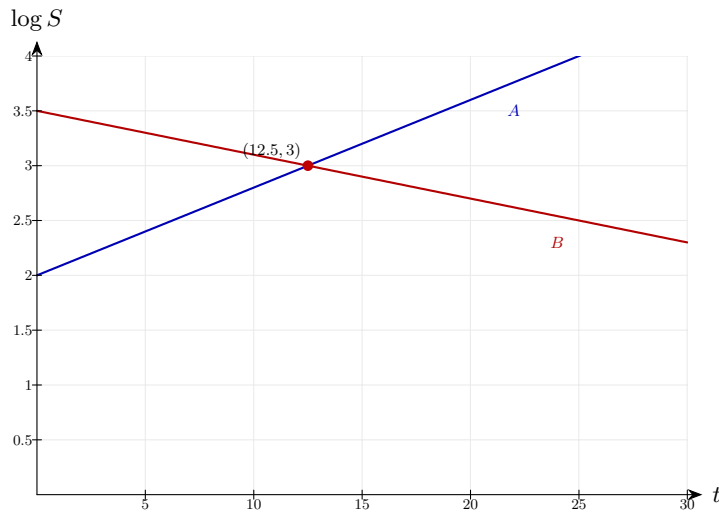
Step 2: (b) Set the two equations equal.

$$0.08t + 2 = -0.04t + 3.5 \Rightarrow 0.12t = 1.5 \Rightarrow t = 12.5$$

Equal $\log S$ values mean equal S values.

Final answer: (a) Blog A: 100 subscribers; Blog B: ≈ 3162 subscribers (b) Day 12.5 (i.e. during day 13)

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 27 (e16-027)

Difficulty: Foundation / Marks: 6

Two animal populations on an island are modelled by $\ln N$ against t (years) graphs:

Deer: $\ln N = 0.15t + 5$

Wolves: $\ln N = 0.25t + 3$

- Find the initial population of each species.
- Find the year t when the wolf population overtakes the deer population.
- Find the combined population at this point. Give your answer to the nearest whole number.

Worked solution.

Step 1: (a) At $t = 0$.

$$\text{Deer: } N = e^5 \approx 148 \quad \text{Wolves: } N = e^3 \approx 20$$

Substitute $t = 0$ into each equation.

Step 2: (b) Set $\ln N$ values equal.

$$0.15t + 5 = 0.25t + 3 \Rightarrow 2 = 0.1t \Rightarrow t = 20$$

The wolves overtake the deer when the lines intersect.

Step 3: (c) Find each population at $t = 20$.

$$\ln N = 0.15(20) + 5 = 8 \Rightarrow N = e^8 \approx 2981$$

Both models give the same N at the intersection.

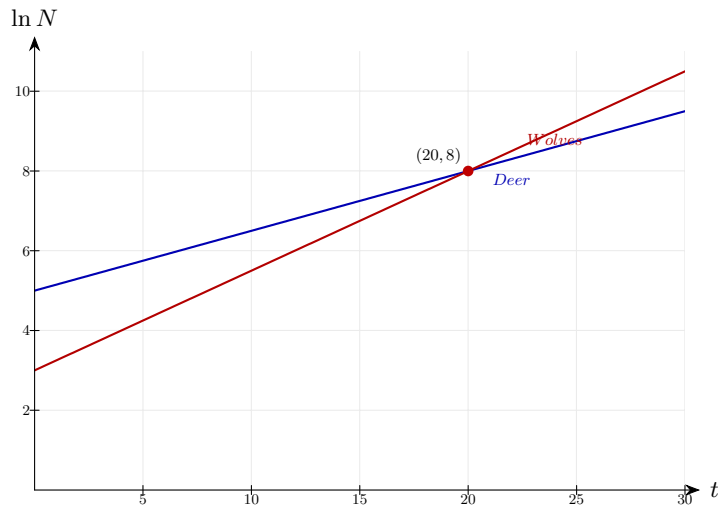
Step 4: Combined population.

$$2 \times e^8 \approx 2 \times 2981 = 5962$$

Both species have the same population at the intersection.

Final answer: (a) Deer ≈ 148 , Wolves ≈ 20 (b) $t = 20$ years (c) ≈ 5962

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 28 (e16-028)

Difficulty: Foundation / Marks: 5

A graph of $\log y$ against $\log x$ is a straight line passing through $(0.4, 1.0)$ and $(1.2, 2.6)$.

(a) Find the equation of the line.

(b) Hence write down the equation relating y and x in the form $y = ax^n$.

(c) Find y when $x = 50$. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Find the gradient.

$$n = \frac{2.6 - 1.0}{1.2 - 0.4} = \frac{1.6}{0.8} = 2$$

Gradient = n in $y = ax^n$.

Step 2: Find the intercept.

$$\log a = 1.0 - 2(0.4) = 1.0 - 0.8 = 0.2$$

Substitute point $(0.4, 1.0)$ into $\log y = n \log x + \log a$.

Step 3: Write the line equation.

$$\log y = 2 \log x + 0.2$$

Straight-line equation in log form.

Step 4: (b) Convert back to original form.

$$a = 10^{0.2} \approx 1.58 \Rightarrow y \approx 1.58x^2$$

Apply $10^{\square}$ to the intercept.

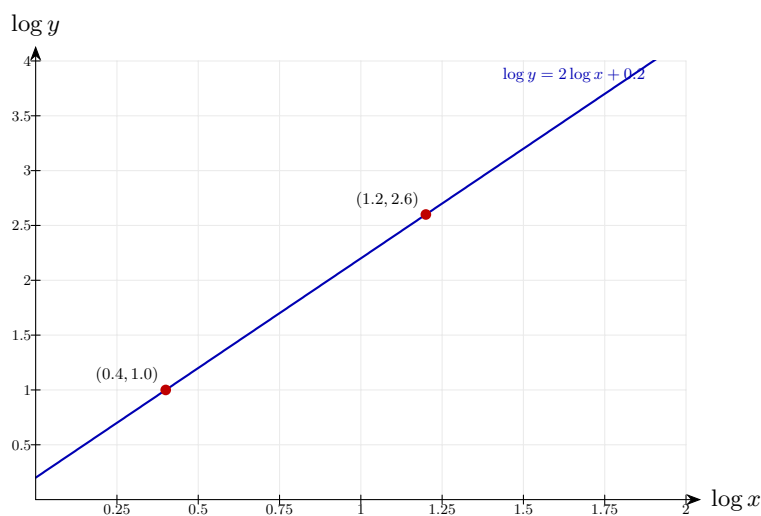
Step 5: (c) Substitute $x = 50$.

$$y \approx 1.58 \times 50^2 = 1.58 \times 2500 = 3950$$

Or use $\log y = 2 \log 50 + 0.2 = 2(1.699) + 0.2 = 3.598 \Rightarrow y = 10^{3.598} \approx 3960$.

Final answer: (a) $\log y = 2 \log x + 0.2$ (b) $y \approx 1.58x^2$ (c) $y \approx 3950$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 29 (e16-029)

Difficulty: Foundation / Marks: 3

A model $y = 10 \times 3^x$ is used to predict the number of users y of a social network after x months.

(a) Write the model in the form $\log y = mx + c$.

(b) Describe one limitation of this model over a long time period.

Worked solution.

Step 1: (a) Take log of both sides.

$$\log y = x \log 3 + \log 10 = (\log 3)x + 1$$

$$\log 10 = 1.$$

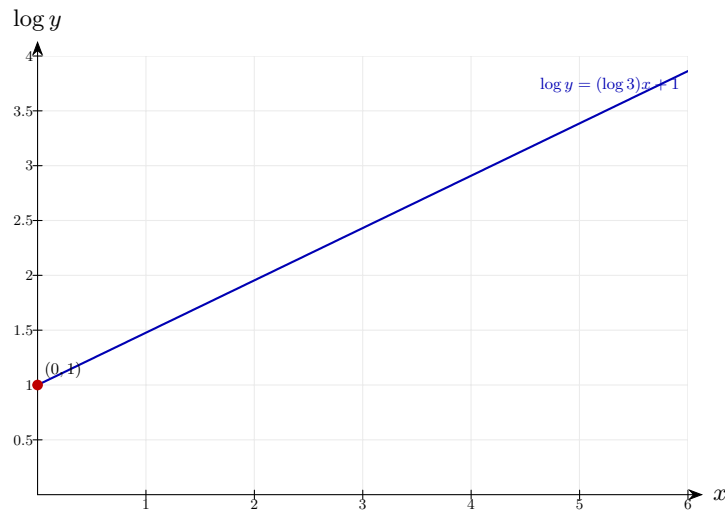
Step 2: (b) Describe the limitation.

$$\text{As } x \rightarrow \infty, y \rightarrow \infty$$

The model predicts unlimited user growth. In reality, the number of users is capped by the world's population and growth slows as the market saturates.

Final answer: (a) $\log y = (\log 3)x + 1$ (b) The model predicts unlimited growth, which is unrealistic — user numbers would saturate once the available population is reached.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 30 (e16-030)

Difficulty: Foundation / Marks: 4

A student plots $\log y$ against $\log x$ for a data set and obtains a straight line. They conclude the data follows $y = ax^n$.

Another student plots $\log y$ against x for the same data and also obtains a straight line. They conclude the data follows $y = kb^x$.

(a) Explain how both students can be correct.

(b) Describe how you would distinguish between the two models experimentally.

Worked solution.

Step 1: (a) Both linearisations can sometimes give approximate straight lines for finite data sets with noise.

$$\text{If } y = ax^n : \log y = n \log x + \log a \quad \text{If } y = kb^x : \log y = x \log b + \log k$$

For limited data ranges, either model might appear linear.

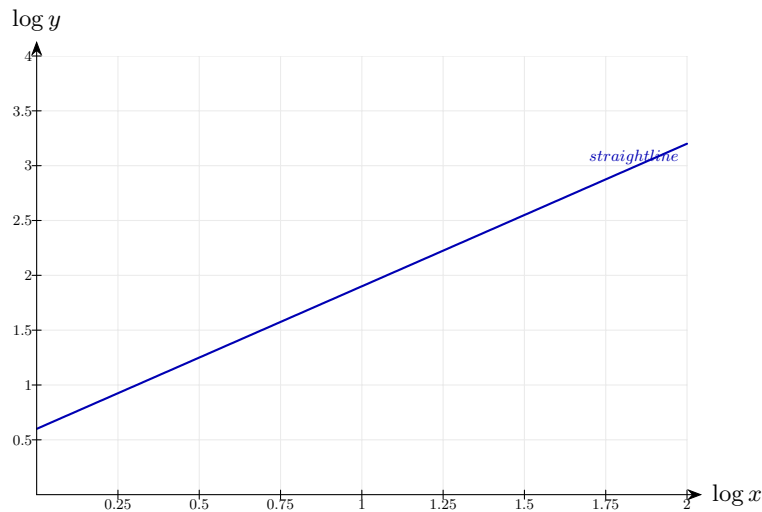
Step 2: (b) Plot both graphs and assess goodness of fit.

Plot $\log y$ vs $\log x$ AND $\log y$ vs x ; the better straight line identifies the model.

The model whose linearised graph fits the data more closely (e.g. has points closer to the line of best fit) is the more appropriate model. The residuals should be examined.

Final answer: (a) With limited data, both transforms may approximate a straight line. (b) Plot both linearised graphs and choose the one whose points lie closer to a straight line of best fit.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 31 (e16-031)

Difficulty: Foundation / Marks: 7

The number of employees p in a company t years after it was founded is modelled by $p = at^b$. The table of values below is given:

t	2	5	10	20	40
p	15	28	45	72	114

- Copy and complete a table of $\log t$ and $\log p$ values to 3 d.p.
- Plot a linear graph and draw a line of best fit.
- Find the values of a and b .
- Predict the number of employees after 60 years.

Worked solution.

Step 1: (a) Compute $\log t$ and $\log p$.

$\log t$	0.301	0.699	1.000	1.301	1.602
$\log p$	1.176	1.447	1.653	1.857	2.057

Take $\log_{10}$ of each value of t and p .

Step 2: (c) Gradient using first and last points.

$$b = \frac{2.057 - 1.176}{1.602 - 0.301} = \frac{0.881}{1.301} \approx 0.677$$

Gradient = b .

Step 3: Find $\log a$.

$$\log a = 1.176 - 0.677(0.301) = 1.176 - 0.204 = 0.972 \Rightarrow a = 10^{0.972} \approx 9.38$$

Substitute the first data point.

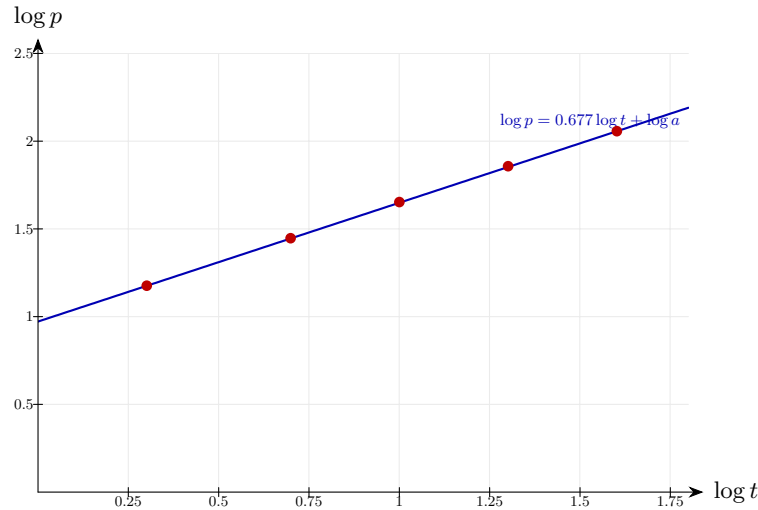
Step 4: (d) Predict at $t = 60$.

$$p = 9.38 \times 60^{0.677} \approx 9.38 \times 15.99 \approx 150$$

$60^{0.677} \approx 15.99$; evaluate using a calculator. Equivalently, $\log p = 0.677 \log 60 + 0.972 = 0.677(1.778) + 0.972 \approx 2.176 \Rightarrow p \approx 10^{2.176} \approx 150$.

Final answer: (c) $b \approx 0.677$, $a \approx 9.38$ (d) ≈ 150 employees

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 32 (e16-032)

Difficulty: Foundation / Marks: 6

The value £ V of a piece of equipment t years old is modelled by $V = pq^t$. A graph of $\log V$ against t has gradient $-\frac{1}{5}$ and passes through $(0, 4)$.

(a) Find p and q to 3 s.f.

(b) Find the age at which the equipment is worth less than £1000.

(c) Write an equation for the linear graph and state the axes that should be plotted.

Worked solution.

Step 1: (a) Vertical intercept gives $\log p$.

$$\log p = 4 \Rightarrow p = 10^4 = 10\,000$$

The intercept of $\log V$ axis.

Step 2: Gradient gives $\log q$.

$$\log q = -0.2 \Rightarrow q = 10^{-0.2} \approx 0.631$$

Gradient = $\log q$.

Step 3: (b) Set $V = 1000$ and use the linear graph.

$$\log(1000) = -0.2t + 4 \Rightarrow 3 = -0.2t + 4 \Rightarrow 0.2t = 1 \Rightarrow t = 5$$

Substitute $\log V = 3$ into the line equation.

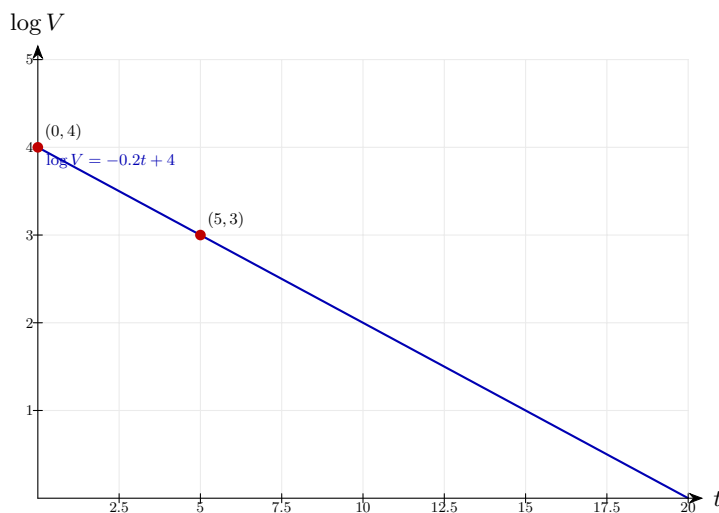
Step 4: (c) The linear equation.

$$\log V = -0.2t + 4 \quad \text{Plot } \log V \text{ against } t$$

Plot $\log_{10} V$ on the vertical axis and t on the horizontal axis.

Final answer: (a) $p = 10\,000$, $q \approx 0.631$ (b) After $t = 5$ years (c) $\log V = -0.2t + 4$; plot $\log V$ against t

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 33 (e16-033)

Difficulty: Foundation / Marks: 7

The number of bacteria p (in thousands) in a petri dish after t days satisfies $p = at^b$.
Observations give:

t	1	3	6	9
p	2	14	44	88

- Plot a linear graph of $\log p$ against $\log t$ using the data.
- Find a and b to 3 s.f.
- Predict the number of bacteria after 15 days.
- Describe one reason why the model may not be appropriate after many days.

Worked solution.

Step 1: (a) Compute $\log t$ and $\log p$.

$\log t$	0	0.477	0.778	0.954
$\log p$	0.301	1.146	1.643	1.944

Take $\log_{10}$ of each value.

Step 2: (b) Gradient using first and last points.

$$b = \frac{1.944 - 0.301}{0.954 - 0} = \frac{1.643}{0.954} \approx 1.72$$

Gradient = b .

Step 3: Find $\log a$ at $\log t = 0$.

$$\log a = 0.301 \Rightarrow a = 10^{0.301} \approx 2.00$$

When $\log t = 0$ (i.e. $t = 1$), $\log p = \log a$.

Step 4: (c) Predict at $t = 15$.

$$p = 2.00 \times 15^{1.72} \approx 2.00 \times 105.4 \approx 211 \text{ thousand}$$

$15^{1.72} \approx 105.4$; evaluate using a calculator. Equivalently, $\log p = 1.72 \log 15 + 0.301 \approx 1.72(1.176) + 0.301 \approx 2.324 \Rightarrow p \approx 10^{2.324} \approx 211 \text{ thousand}$.

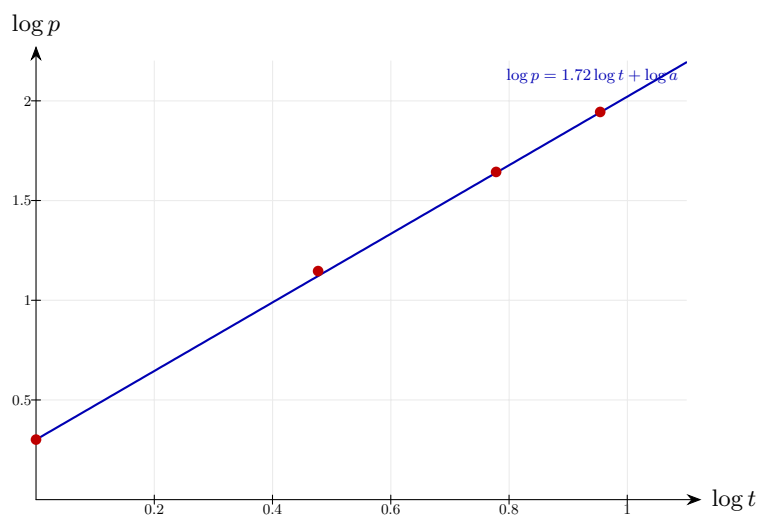
Step 5: (d) Describe a limitation.

$$p \rightarrow \infty \text{ as } t \rightarrow \infty$$

Nutrients in the petri dish would eventually be exhausted, limiting growth. The model ignores resource constraints.

Final answer: (b) $b \approx 1.72$, $a \approx 2.00$ (c) $\approx 211\,000$ bacteria (d) Nutrient depletion would slow/stop growth — the model allows unlimited increase.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 34 (e16-034)

Difficulty: Foundation / Marks: 6

A scientist measures the speed v (m/s) of a chemical reaction at temperature T (K). She believes $v = aT^n$. A graph of $\log v$ against $\log T$ gives a straight line with equation $\log v = 2.5 \log T - 5.4$.

- (a) Find a and n to 3 s.f.
- (b) Find v when $T = 400$ K.
- (c) Find T when $v = 80$. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Read off n and find a .

$$n = 2.5 \quad \log a = -5.4 \Rightarrow a = 10^{-5.4} \approx 3.98 \times 10^{-6}$$

Gradient = n ; intercept = $\log a$.

Step 2: (b) Substitute $T = 400$.

$$\log v = 2.5 \log(400) - 5.4 = 2.5(2.602) - 5.4 = 6.505 - 5.4 = 1.105 \Rightarrow v = 10^{1.105} \approx 12.7 \text{ m/s}$$

$$\log 400 \approx 2.602.$$

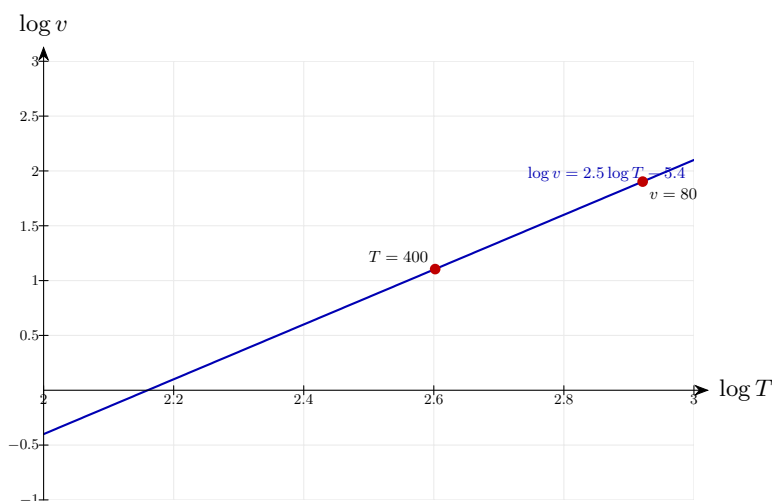
Step 3: (c) Set $v = 80$ and solve.

$$\log 80 = 2.5 \log T - 5.4 \Rightarrow 1.903 = 2.5 \log T - 5.4 \Rightarrow 2.5 \log T = 7.303 \Rightarrow \log T = 2.921 \Rightarrow T = 10^{2.921} \approx 834 \text{ K}$$

$\log 80 \approx 1.903$. Solve for $\log T$ then apply $10^{\square}$.

Final answer: (a) $n = 2.5$, $a \approx 3.98 \times 10^{-6}$ (b) $v \approx 12.7 \text{ m/s}$ (c) $T \approx 834 \text{ K}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 35 (e16-035)

Difficulty: Foundation / Marks: 7

The populations (in thousands) of rabbits R and foxes F on an island are modelled by straight-line graphs of $\ln N$ against t (years):

Rabbits: $\ln R = -0.08t + 8$

Foxes: $\ln F = 0.12t + 4$

- Find the initial population of each species. Give your answers to 3 s.f.
- Find the year t when both populations are equal.
- Find the combined population at this time. Give your answer to 3 s.f.
- Explain why the rabbit model may not be appropriate for large t .

Worked solution.

Step 1: (a) Initial populations at $t = 0$.

$$\ln R = 8 \Rightarrow R = e^8 \approx 2981 \text{ thousand} \quad \ln F = 4 \Rightarrow F = e^4 \approx 54.6 \text{ thousand}$$

Substitute $t = 0$ into each equation.

Step 2: (b) Set the two $\ln N$ expressions equal.

$$-0.08t + 8 = 0.12t + 4 \Rightarrow 4 = 0.2t \Rightarrow t = 20$$

Equal $\ln N$ values imply equal N values.

Step 3: (c) Find the common population at $t = 20$.

$$\ln N = -0.08(20) + 8 = -1.6 + 8 = 6.4 \Rightarrow N = e^{6.4} \approx 601 \text{ thousand}$$

Substitute $t = 20$ into either equation.

Step 4: Combined population.

$$2 \times 601 \approx 1202 \text{ thousand} \approx 1\,200\,000$$

Both populations are equal at this point.

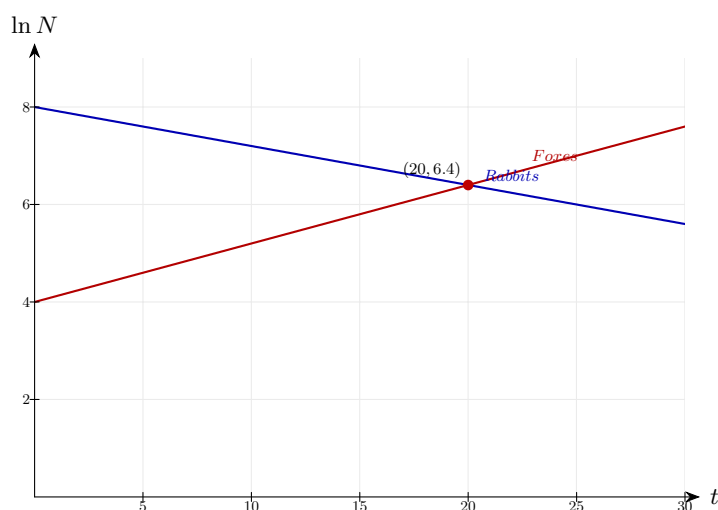
Step 5: (d) Limitation of the rabbit model.

$$\text{As } t \rightarrow \infty, \ln R \rightarrow -\infty, \text{ so } R \rightarrow 0$$

The model predicts the rabbit population tends to zero. In reality, if foxes decline too, the rabbits might recover. The model ignores predator–prey interactions and other ecological factors.

Final answer: (a) Rabbits $\approx 2\,981\,000$; Foxes $\approx 54\,600$ (b) $t = 20$ years (c) $\approx 1\,202\,000$
 (d) The model predicts $R \rightarrow 0$, which may be unrealistic — it ignores ecological interactions that could allow recovery.

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 36 (e16-036)

Difficulty: Foundation / Marks: 3

A graph of $\log y$ against $\log x$ is a straight line with gradient 2 and y-intercept 0.5. Write down the equation connecting y and x .

Worked solution.

Step 1: Line equation

$$\log y = 2 \log x + 0.5$$

A $\log y$ vs $\log x$ plot has the form $\log y = n \log x + \log a$, so read the gradient as n and the vertical intercept as $\log a$.

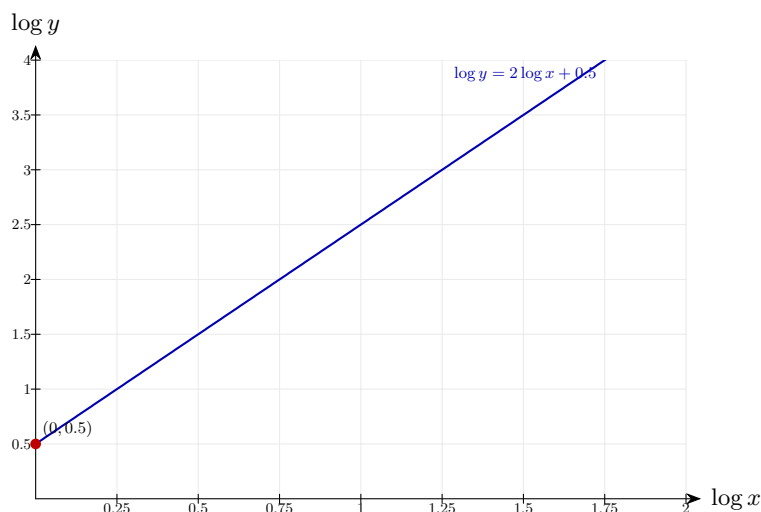
Step 2: Convert

$$\log y = \log x^2 + \log 10^{0.5} \implies y = \sqrt{10} \cdot x^2 \approx 3.16x^2$$

Apply $10^{\square}$ to both sides: the power law turns $2 \log x$ into $\log x^2$, and the constant 0.5 becomes $10^{0.5} = \sqrt{10}$. So $a = 10^{0.5} \approx 3.16$ and $n = 2$.

Final answer: $y = \sqrt{10} \cdot x^2$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 37 (e16-037)

Difficulty: Foundation / Marks: 3

A graph of $\log y$ against x is a straight line with gradient 0.2 and y-intercept 1. Find k and b in $y = kb^x$.

Worked solution.

Step 1: Line equation

$$\log y = 0.2x + 1$$

For $y = kb^x$, taking $\log_{10}$ gives $\log y = (\log b)x + \log k$. Note this is $\log y$ vs x (not $\log x$) — so gradient is $\log b$ and intercept is $\log k$.

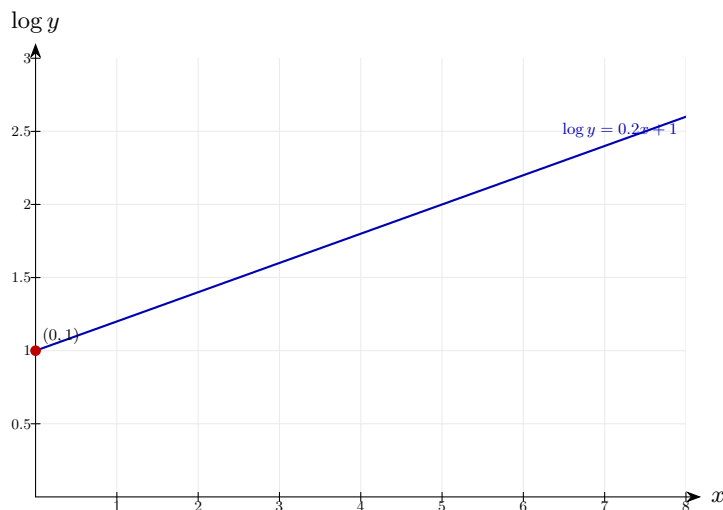
Step 2: Convert

$$k = 10^1 = 10; \quad b = 10^{0.2} \approx 1.585$$

Recover k and b by applying $10^\square$ to the intercept and gradient respectively.

Final answer: $k = 10, b = 1.585$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 38 (e16-038)

Difficulty: Foundation / Marks: 3

Explain why plotting $\log y$ against $\log x$ for data following $y = ax^n$ gives a straight line.

Worked solution.

Step 1: Take log of both sides

$$\log y = \log(ax^n) = \log a + n \log x$$

Use the product law $\log(ax^n) = \log a + \log(x^n)$, then the power law $\log(x^n) = n \log x$. The variable x is "trapped" inside an exponent, but taking logs brings it down as a coefficient.

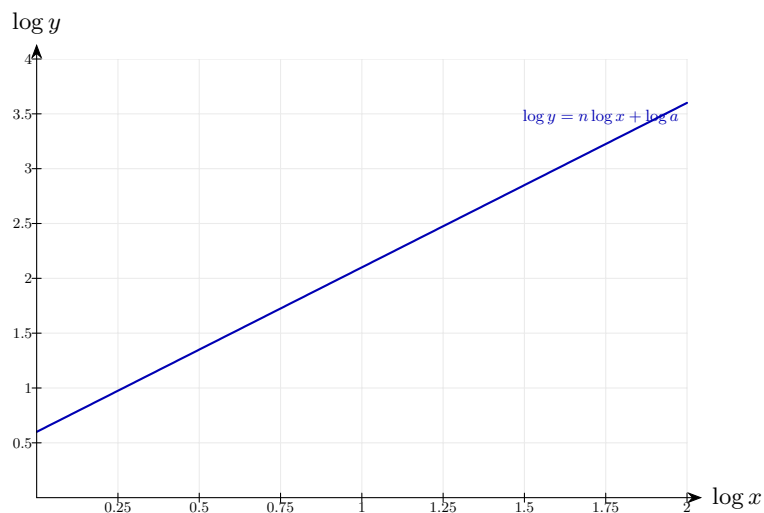
Step 2: This is linear

$$\log y = n \log x + \log a$$

Compare with $Y = mX + c$ where $Y = \log y$, $X = \log x$, $m = n$, $c = \log a$. Since a and n are constants, both m and c are constants — so the plot of $\log y$ against $\log x$ is a straight line.

Final answer: Linear form: $\log y = n \log x + \log a$; gradient n , intercept $\log a$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 39 (e16-039)

Difficulty: Foundation / Marks: 3

Explain why plotting $\log y$ against x for data following $y = kb^x$ gives a straight line.

Worked solution.

Step 1: Take log

$$\log y = \log k + x \log b$$

Apply $\log_{10}$ to both sides of $y = kb^x$, then use the product and power laws. The crucial step: $\log(b^x) = x \log b$ — this brings x out of the exponent, making it appear linearly.

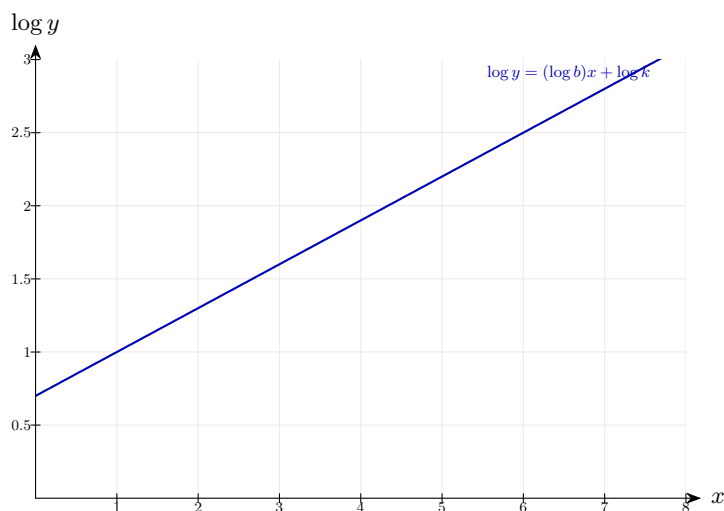
Step 2: Linear in x

$$\text{Gradient} = \log b, \quad \text{intercept} = \log k$$

Note the axes: this is $\log y$ against x , not $\log x$. The gradient is $\log b$ (not b itself) and the intercept is $\log k$.

Final answer: Linear form: $\log y = (\log b)x + \log k$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 40 (e16-040)

Difficulty: Foundation / Marks: 3

A graph of $\ln y$ against x has gradient 0.5 and passes through (0, 2). Find the equation in the form $y = ke^{cx}$.

Worked solution.

Step 1: Line equation

$$\ln y = 0.5x + 2$$

For $y = ke^{cx}$, taking natural logs gives $\ln y = cx + \ln k$; so gradient = c and intercept = $\ln k$. Here $\ln$ is consistent throughout because the model uses base e .

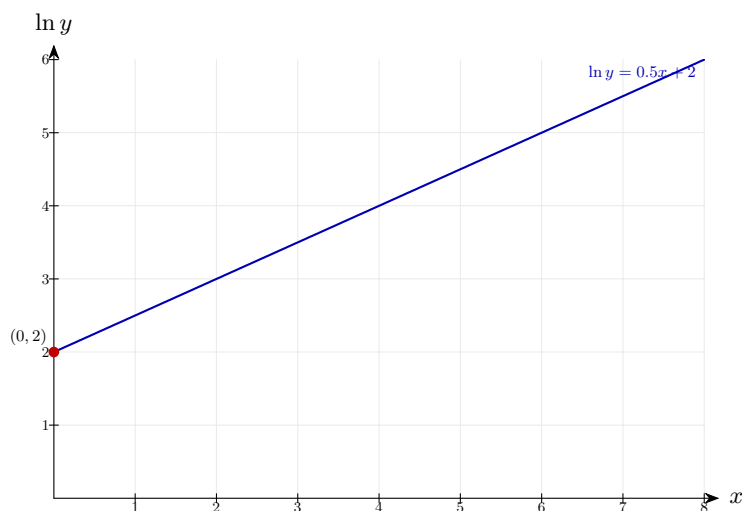
Step 2: Exponentiate

$$y = e^{0.5x+2} = e^2 \cdot e^{0.5x} \approx 7.39e^{0.5x}$$

Apply $e^{\square}$ to both sides, then split using $e^{a+b} = e^a e^b$. So $k = e^2 \approx 7.39$ and $c = 0.5$.

Final answer: $y = e^2 \cdot e^{0.5x}$; $k = e^2$, $c = 0.5$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 41 (e16-041)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against $\log x$ passes through (0.5, 1.2) and (2.0, 3.0). Find a and n in $y = ax^n$.

Worked solution.

Step 1: Gradient

$$n = \frac{3.0 - 1.2}{2.0 - 0.5} = \frac{1.8}{1.5} = 1.2$$

Gradient = $\Delta(\log y)/\Delta(\log x)$, which equals n in the linearised form $\log y = n \log x + \log a$.

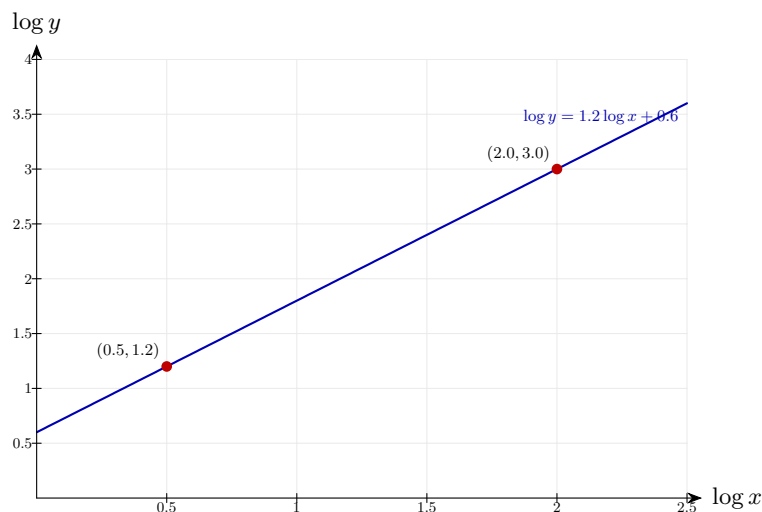
Step 2: Find $\log a$

$$1.2 = 1.2(0.5) + \log a \implies \log a = 0.6 \implies a = 10^{0.6} = 3.98$$

Substitute either point into $\log y = n \log x + \log a$ and solve for $\log a$, then apply $10^{\square}$ to recover a .

Final answer: $n = 1.2, a = 3.98$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 42 (e16-042)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against x passes through $(0, 1.5)$ and $(4, 3.5)$. Find k and b in $y = kb^x$.

Worked solution.

Step 1: Gradient

$$\log b = \frac{3.5 - 1.5}{4} = 0.5 \implies b = 10^{0.5} = \sqrt{10} \approx 3.16$$

For $y = kb^x$, gradient = $\log b$ — be careful: the gradient is not b itself. Apply $10^{\square}$ to recover b .

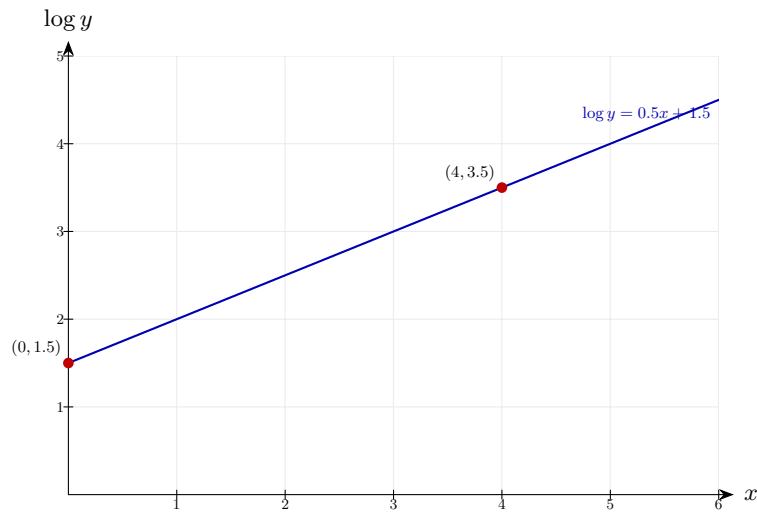
Step 2: k from intercept

$$k = 10^{1.5} \approx 31.6$$

Since one of the points is $(0, 1.5)$, this is directly on the $\log y$ -axis — so the intercept = $\log k = 1.5$, giving $k = 10^{1.5}$.

Final answer: $k = 31.6, b = 3.16$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 43 (e16-043)

Difficulty: Foundation / Marks: 4

Data: $x = 2, y = 20$; $x = 5, y = 125$. **Verify these satisfy $y = ax^n$ and find a and n .**

Worked solution.

Step 1: Take logs

$$\log 20 = 1.301, \log 2 = 0.301; \quad \log 125 = 2.097, \log 5 = 0.699$$

Convert each data point to $(\log x, \log y)$ form so we can work with the linearised equation $\log y = n \log x + \log a$.

Step 2: Gradient

$$n = \frac{2.097 - 1.301}{0.699 - 0.301} = \frac{0.796}{0.398} = 2.0$$

Compute the gradient of the $\log y$ vs $\log x$ line through the two points — this equals n .

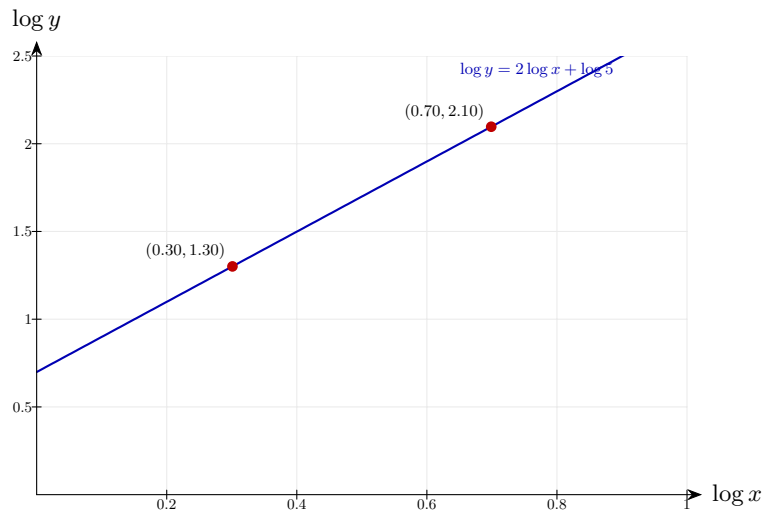
Step 3: Find a

$$\log a = 1.301 - 2(0.301) = 0.699 \implies a = 5$$

Substitute the first point into $\log y = n \log x + \log a$ and solve. Quick check: $y = 5x^2$ gives $5(2)^2 = 20$ and $5(5)^2 = 125$ — both match.

Final answer: $y = 5x^2$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 44 (e16-044)

Difficulty: Foundation / Marks: 4

Data: $x = 1, y = 6$; $x = 3, y = 54$. These satisfy $y = kb^x$. Find k and b .

Worked solution.

Step 1: Set up equations

$$6 = kb; \quad 54 = kb^3$$

Substitute each data point into $y = kb^x$ to get two simultaneous equations.

Step 2: Divide

$$\frac{54}{6} = b^2 \implies b = 3$$

Dividing the equations cancels k , leaving a single equation in b . Take the positive root since $b > 0$ for the model to make sense.

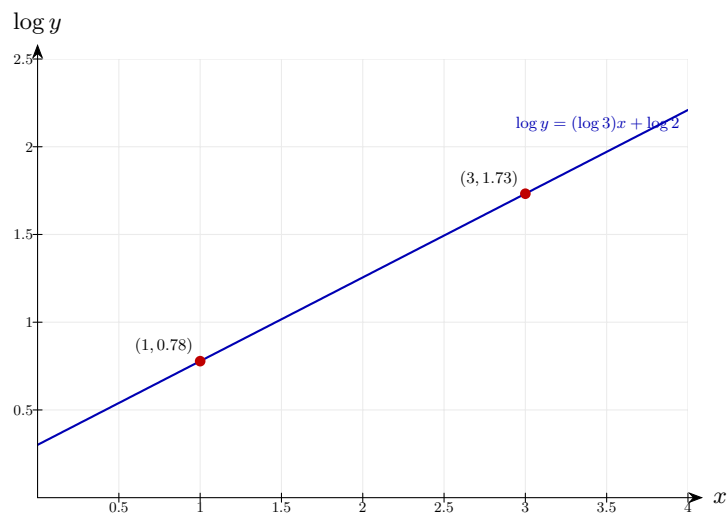
Step 3: Find k

$$k = \frac{6}{3} = 2$$

Substitute $b = 3$ back into the first equation to find k .

Final answer: $k = 2, b = 3; y = 2 \times 3^x$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 45 (e16-045)

Difficulty: Foundation / Marks: 3

The equation $\log y = 0.3x + 1.2$ relates x and y . Find y when $x = 5$.

Worked solution.

Step 1: Substitute

$$\log y = 0.3(5) + 1.2 = 2.7$$

Direct substitution into the linearised equation gives $\log y$. We have not yet got y itself — only its $\log$.

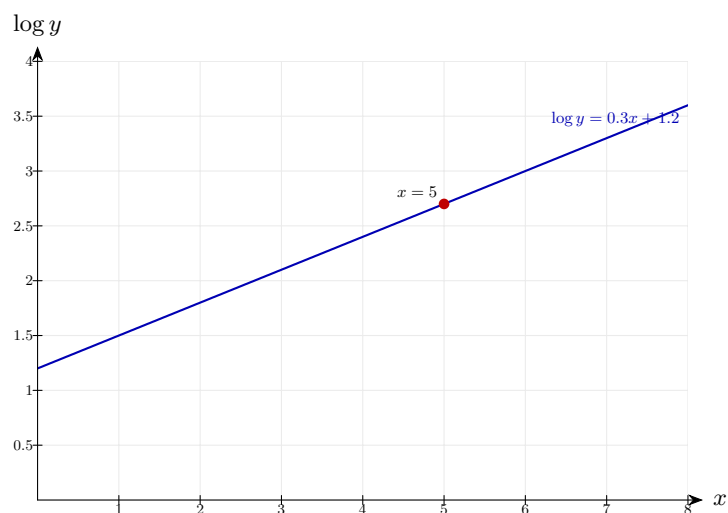
Step 2: Find y

$$y = 10^{2.7} = 501$$

Undo the $\log_{10}$ by applying $10^{\square}$ to both sides.

Final answer: $y = 501$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 46 (e16-046)

Difficulty: Foundation / Marks: 3

The equation $\log y = 1.5 \log x + 0.8$ relates x and y . Find y when $x = 4$.

Worked solution.

Step 1: Substitute

$$\log y = 1.5 \log 4 + 0.8 = 1.5(0.602) + 0.8 = 1.703$$

Plug $x = 4$ into the equation, using $\log 4 \approx 0.602$. The result is $\log y$, not y .

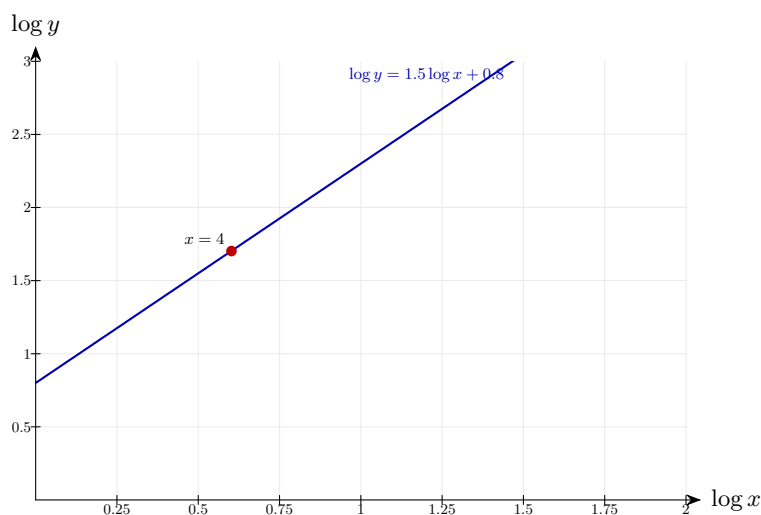
Step 2: Find y

$$y = 10^{1.703} = 50.4$$

Apply $10^{\square}$ to recover y .

Final answer: $y = 50.4$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 47 (e16-047)

Difficulty: Foundation / Marks: 2

Given $\ln y = 2x + 3$, express y in the form $y = Ae^{Bx}$.

Worked solution.

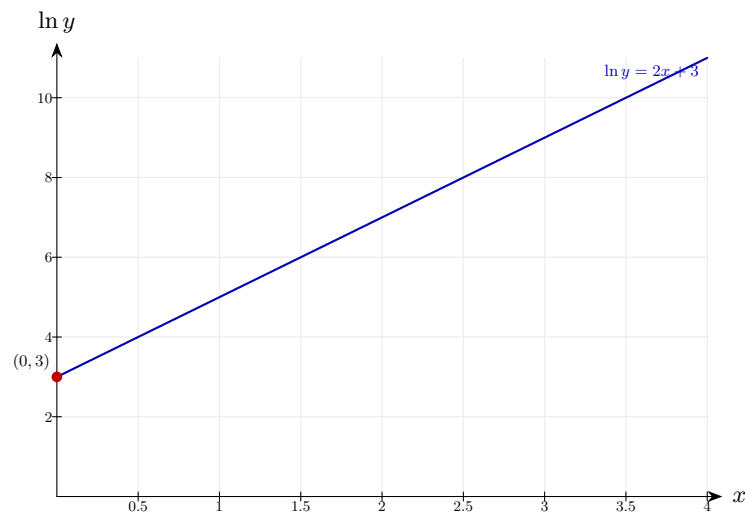
Exponentiate

$$y = e^{2x+3} = e^3 \cdot e^{2x}$$

Apply $e^{\square}$ to both sides (the inverse of $\ln$), then split using $e^{a+b} = e^a e^b$. This separates the constant factor $A = e^3$ from the exponential factor e^{Bx} .

Final answer: $y = e^3 e^{2x}$; $A = e^3$, $B = 2$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 48 (e16-048)

Difficulty: Foundation / Marks: 3

A straight line on a $\log y$ vs $\log x$ graph has equation $\log y = -2\log x + 3$. Write y as a function of x .

Worked solution.

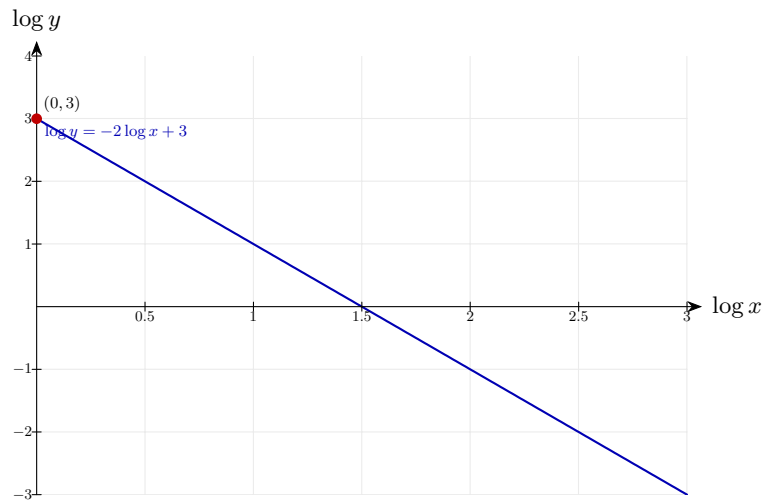
Convert

$$\log y = \log x^{-2} + \log 10^3 = \log \frac{1000}{x^2}$$

Use the power law to rewrite $-2\log x$ as $\log x^{-2}$, and 3 as $\log 10^3$. Then combine using the product law. A negative gradient on $\log y$ vs $\log x$ means y decreases as x increases.

Final answer: $y = 1000/x^2$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 49 (e16-049)

Difficulty: Foundation / Marks: 3

A relationship $y = 3 \times 2^x$ is plotted as $\log y$ against x . Find the gradient and y-intercept of the resulting line.

Worked solution.

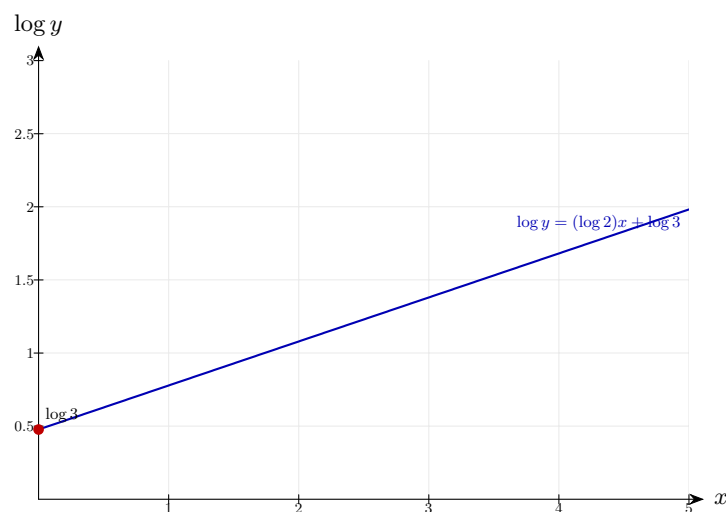
Take $\log$

$$\log y = \log 3 + x \log 2 = 0.477 + 0.301x$$

Apply $\log_{10}$ and use the product/power laws. Rearrange to $\log y = (\log 2)x + \log 3$: the gradient is $\log 2 \approx 0.301$ (not 2) and the intercept is $\log 3 \approx 0.477$ (not 3).

Final answer: Gradient = 0.301, y-intercept = 0.477

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 50 (e16-050)

Difficulty: Foundation / Marks: 3

A relationship $y = 5x^3$ is plotted as $\log y$ against $\log x$. Find the gradient and y-intercept.

Worked solution.

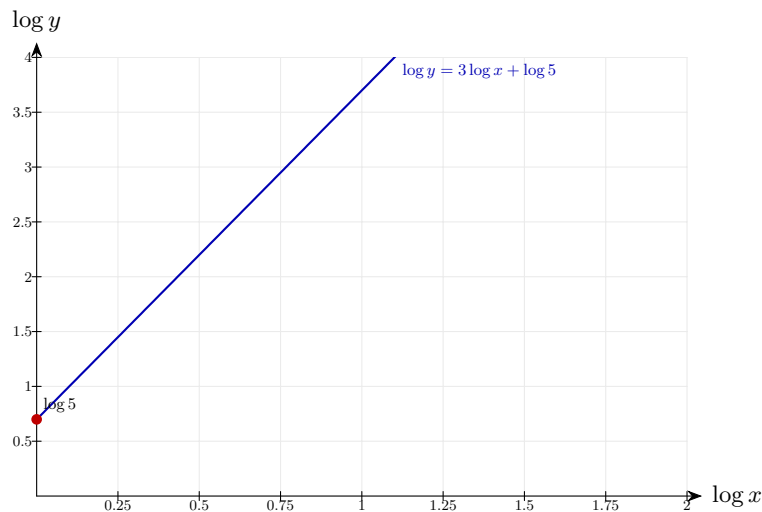
Take $\log$

$$\log y = 3 \log x + \log 5 = 3 \log x + 0.699$$

For a power model $y = ax^n$, the gradient is exactly $n = 3$ (no $\log$ needed) because $\log(x^3) = 3 \log x$. The intercept is $\log a = \log 5 \approx 0.699$.

Final answer: Gradient = 3, y-intercept = 0.699

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 51 (e16-051)

Difficulty: Foundation / Marks: 3

A student plots $\log y$ against $\log x$ and gets gradient 0.5 and intercept 1. What is the relationship?

Worked solution.

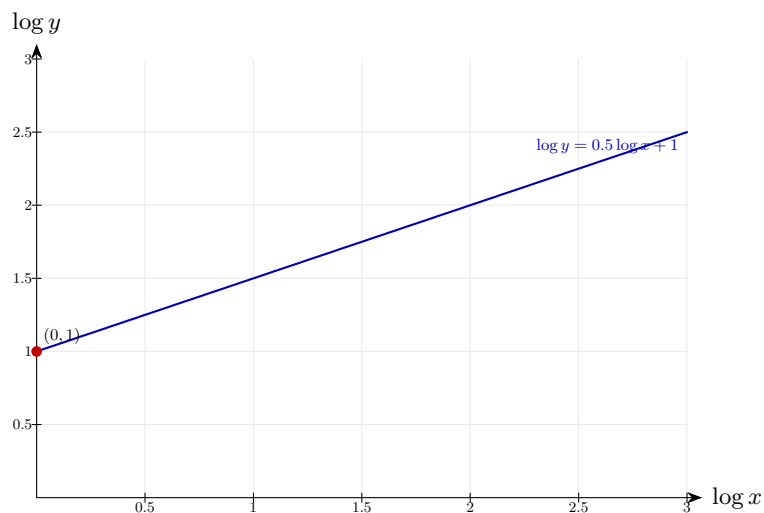
Form equation

$$\log y = 0.5 \log x + 1 \implies y = 10 \cdot x^{0.5} = 10\sqrt{x}$$

A straight line on $\log y$ vs $\log x$ signals a power model $y = ax^n$ with $n = \text{gradient} = 0.5$ and $a = 10^{\text{intercept}} = 10^1 = 10$. Note $x^{0.5} = \sqrt{x}$.

Final answer: $y = 10 \sqrt{x}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 52 (e16-052)

Difficulty: Foundation / Marks: 2

A student plots $\log y$ against x and gets a straight line. Another plots $\log y$ against $\log x$ and gets a curve. What type of relationship does the data follow?

Worked solution.

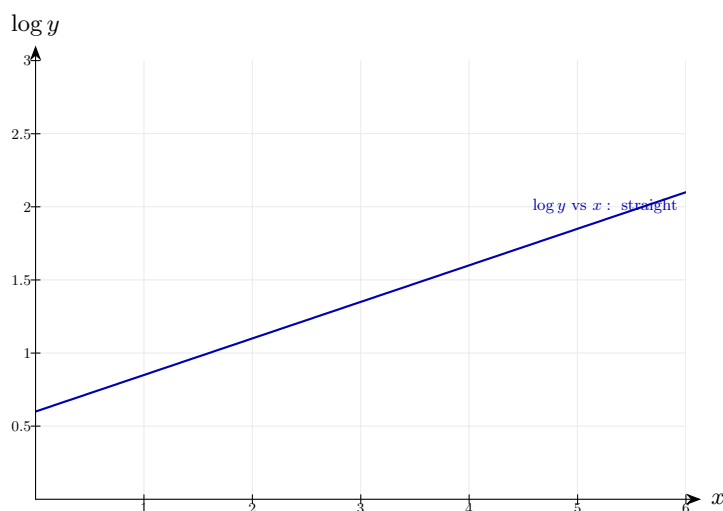
Analyse

$$y = kb^x$$

A straight line on $\log y$ vs x is the signature of an exponential model $y = kb^x$; a straight line on $\log y$ vs $\log x$ is the signature of a power model $y = ax^n$. The first axis combination is linear here, the second is not — so the data is exponential, not power.

Final answer: Exponential: $y = kb^x$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 53 (e16-053)

Difficulty: Foundation / Marks: 4

Given the line $\log y = -0.5 \log x + 2$, find x when $y = 50$.

Worked solution.

Step 1: Substitute

$$\log 50 = -0.5 \log x + 2 \implies 1.699 = -0.5 \log x + 2$$

Replace $\log y$ with $\log 50 \approx 1.699$. The equation becomes linear in $\log x$, which is much easier than solving for x directly.

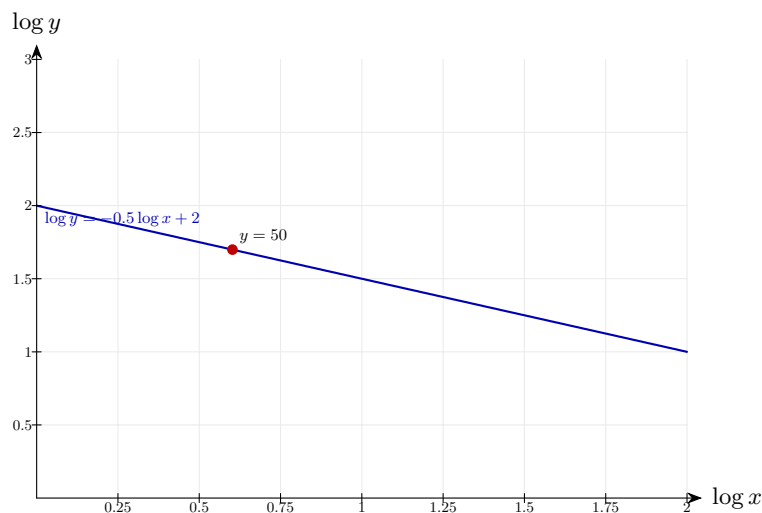
Step 2: Solve

$$\log x = \frac{2 - 1.699}{0.5} = 0.602 \implies x = 4$$

Rearrange to isolate $\log x$, then apply $10^{\square}$ to recover x . Note $10^{0.602} \approx 4$ since $\log 4 \approx 0.602$.

Final answer: $x = 4$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 54 (e16-054)

Difficulty: Foundation / Marks: 3

Given $\log y = 0.4x + 0.8$, find x when $y = 1000$.

Worked solution.

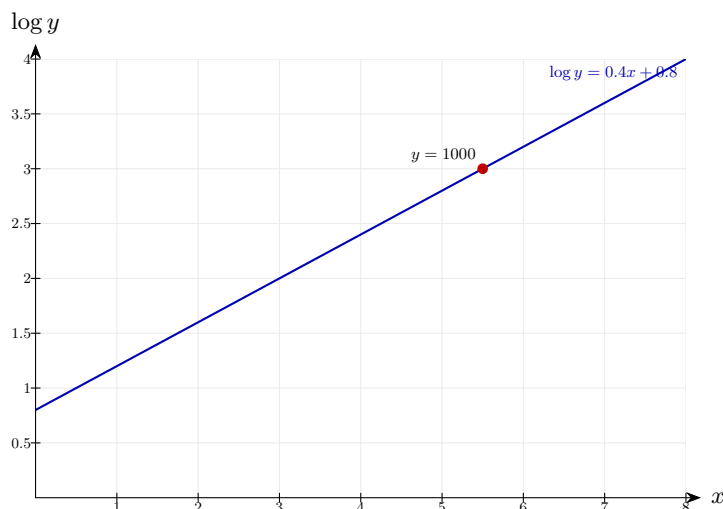
Substitute

$$3 = 0.4x + 0.8 \implies x = \frac{2.2}{0.4} = 5.5$$

$\log 1000 = 3$ because $10^3 = 1000$. Substitute and solve the linear equation for x directly — no need to convert back to the original exponential form.

Final answer: $x = 5.5$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 55 (e16-055)

Difficulty: Foundation / Marks: 3

The speed v and distance d of planets satisfy $v = kd^n$. A plot of $\log v$ against $\log d$ gives gradient -0.5 and intercept 2.1 . Find k and n .

Worked solution.

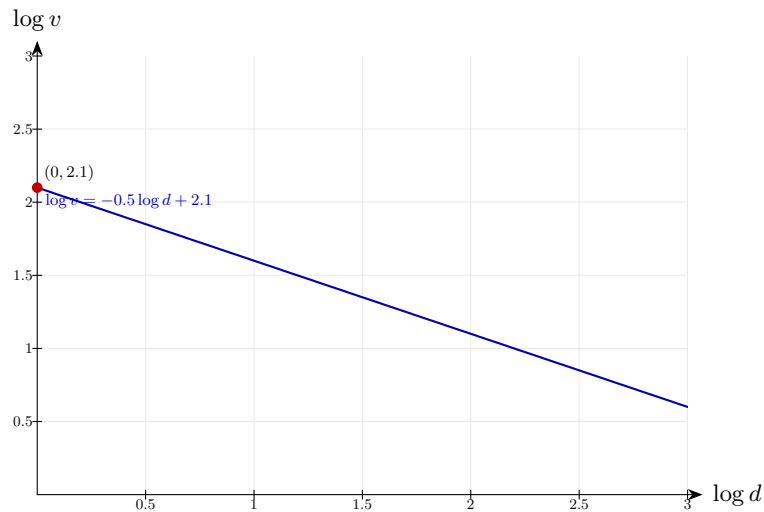
Read off

$$n = -0.5; \quad k = 10^{2.1} = 126$$

For a power model on $\log v$ vs $\log d$ axes, gradient $= n$ directly, and the intercept is $\log k$ — so apply $10^{\square}$ to the intercept to recover k . The negative n here means v decreases as d increases (consistent with Kepler-style planetary motion).

Final answer: $n = -0.5, k = 126$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 56 (e16-056)

Difficulty: Foundation / Marks: 2

A relationship $y = 4e^{-0.3x}$ is plotted as $\ln y$ against x . State the gradient and intercept.

Worked solution.

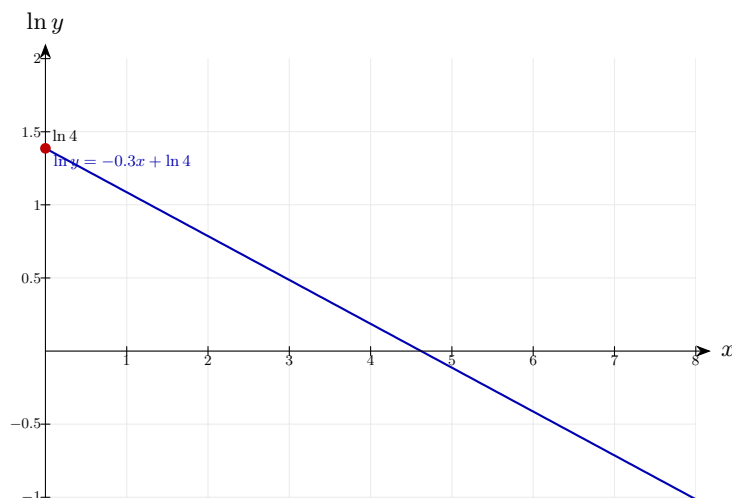
Take $\ln$

$$\ln y = \ln 4 - 0.3x = -0.3x + 1.386$$

Because the model uses base e , $\ln$ gives a particularly clean linearisation: $\ln(e^{-0.3x}) = -0.3x$ directly. The gradient is just the coefficient of x in the exponent, and the intercept is $\ln k$.

Final answer: Gradient = -0.3, intercept = $\ln 4 = 1.386$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 57 (e16-057)

Difficulty: Foundation / Marks: 4

Two points on a $\ln y$ vs x graph are $(2, 3.4)$ and $(8, 6.4)$. Find k and c in $y = ke^{cx}$.

Worked solution.

Step 1: Gradient c

$$c = \frac{6.4 - 3.4}{8 - 2} = 0.5$$

For $y = ke^{cx}$, the linearised form is $\ln y = cx + \ln k$, so the gradient of $\ln y$ vs x equals c directly (no further log/exp needed).

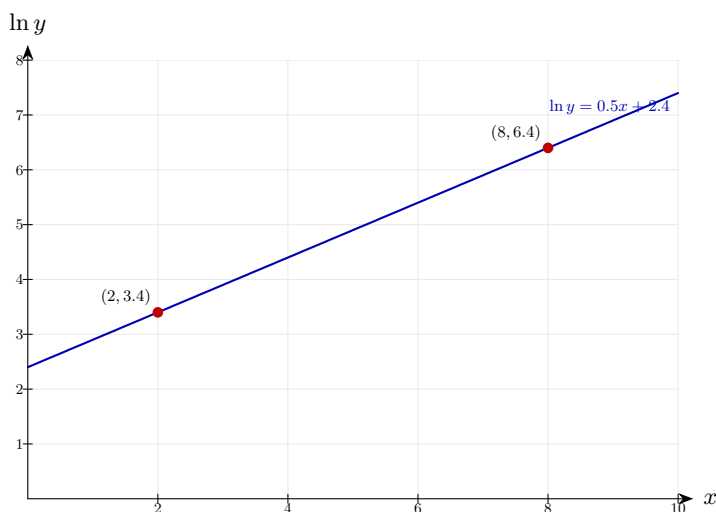
Step 2: Find $\ln k$

$$3.4 = 0.5(2) + \ln k \implies \ln k = 2.4 \implies k = e^{2.4} = 11.0$$

Substitute either point into the linearised equation, solve for $\ln k$, then apply $e^{\square}$ (not $10^{\square}$, because we used $\ln$).

Final answer: $k = 11.0, c = 0.5$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 58 (e16-058)

Difficulty: Foundation / Marks: 4

A decay process has data: $t = 0, y = 100$; $t = 10, y = 37$. Find the equation $y = Ae^{-kt}$.

Worked solution.

Step 1: $A = 100$ (from $t=0$)

$$A = 100$$

At $t = 0$, $e^{-k(0)} = e^0 = 1$, so $y(0) = A$. The initial value gives A directly.

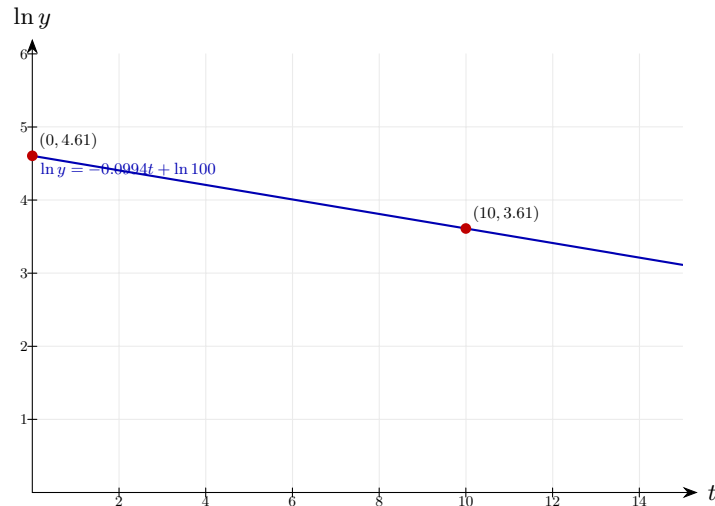
Step 2: Find k

$$37 = 100e^{-10k} \implies k = \frac{-\ln 0.37}{10} = 0.0994$$

Divide both sides by 100, then take $\ln$ of both sides to bring the exponent down: $-10k = \ln 0.37$. Solve for k ; the negative sign of the original exponent absorbs the negative log to give a positive k .

Final answer: $y = 100e^{-0.0994t}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 59 (e16-059)

Difficulty: Foundation / Marks: 3

Which graph type would you plot to test if data follows: (a) $y = ax^n$; (b) $y = kb^x$; (c) $y = ke^{cx}$?

Worked solution.

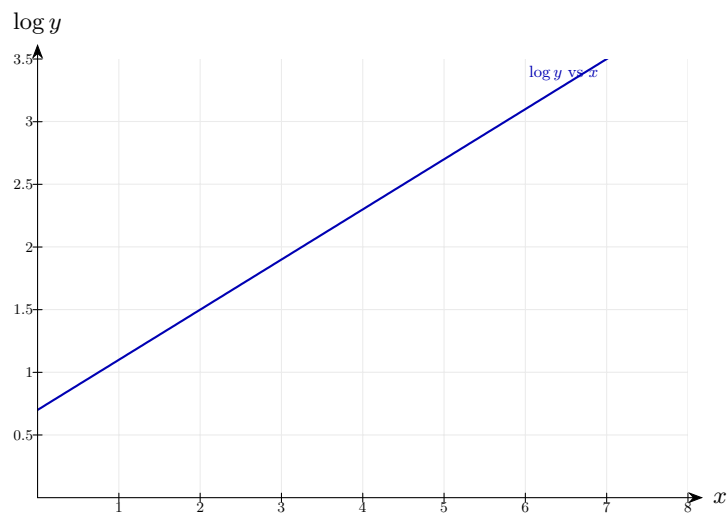
Identify

(a) $\log y$ vs $\log x$; (b) $\log y$ vs x ; (c) $\ln y$ vs x

For a power model, log both variables (x is in the base, so its log must appear). For an exponential model, only $\log y$ (x is in the exponent, so it comes down as a coefficient). Use $\ln$ when the base is e — it cancels cleanly with e^{cx} .

Final answer: (a) $\log y$ vs $\log x$; (b) $\log y$ vs x ; (c) $\ln y$ vs x

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 60 (e16-060)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against $\log x$ gives gradient 3 and passes through $(1, 2)$. Find the equation and evaluate y when $x = 10$.

Worked solution.

Step 1: Equation

$$\log y = 3 \log x + c; \quad 2 = 3(1) + c \implies c = -1$$

Plug the known point $(\log x, \log y) = (1, 2)$ into $\log y = 3 \log x + c$ and solve for the intercept $c = \log a$. Here $\log a = -1$, so $a = 10^{-1} = 0.1$.

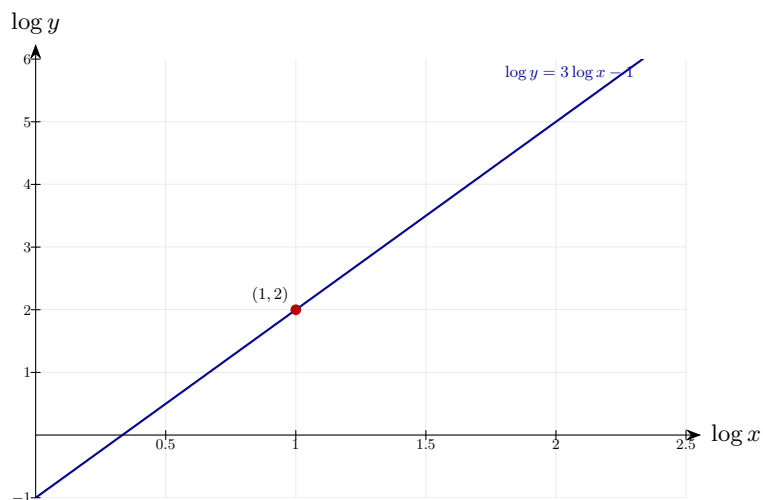
Step 2: So $y = 0.1x^3$

$$x = 10 : y = 0.1 \times 1000 = 100$$

Substitute back into $y = 0.1x^3$. At $x = 10$, $x^3 = 1000$, so $y = 100$.

Final answer: $y = 0.1x^3$; $y(10) = 100$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 61 (e16-061)

Difficulty: Foundation / Marks: 2

Convert $y = 7 \times 5^x$ into the form $\log y = mx + c$ and state m and c .

Worked solution.

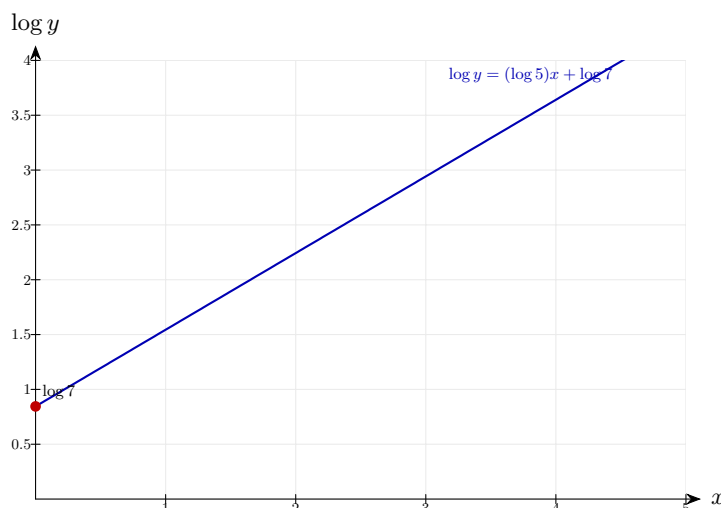
Take log

$$\log y = x \log 5 + \log 7 = 0.699x + 0.845$$

Apply $\log_{10}$ and split using product/power laws. The coefficient of x becomes $m = \log 5 \approx 0.699$ (not 5) — a common slip is forgetting to log the base of the exponential.

Final answer: $m = 0.699, c = 0.845$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 62 (e16-062)

Difficulty: Foundation / Marks: 2

Convert $y = 8x^{2.5}$ into the form $\log y = n \log x + \log a$ and state n and $\log a$.

Worked solution.

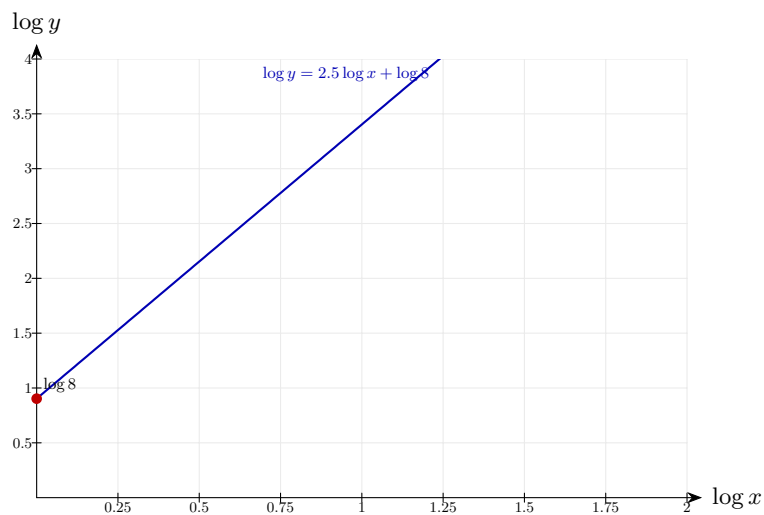
Take log

$$\log y = 2.5 \log x + \log 8 = 2.5 \log x + 0.903$$

For a power model, the power n appears directly as the gradient (no log applied to it) — contrast with the exponential case where the base needs logging. Here $n = 2.5$ and $\log a = \log 8 \approx 0.903$.

Final answer: $n = 2.5, \log a = 0.903$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 63 (e16-063)

Difficulty: Foundation / Marks: 3

The line $\log y = 2 \log x + 0.3$ intersects the line $\log y = 3$. Find the value of x at the intersection.

Worked solution.

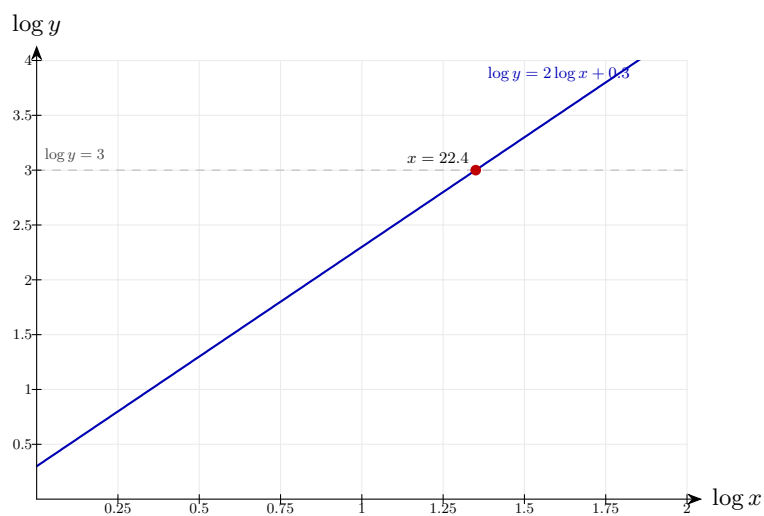
Set equal

$$3 = 2 \log x + 0.3 \implies \log x = 1.35 \implies x = 22.4$$

Equate the two expressions for $\log y$, solve the resulting linear equation for $\log x$, then apply $10^{\square}$ to recover x .

Final answer: $x = 22.4$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 64 (e16-064)

Difficulty: Foundation / Marks: 3

A graph of $\log P$ against $\log V$ for a gas gives a straight line with gradient -1.4 and intercept 2.5 . Find P when $V = 10$.

Worked solution.

Step 1: Equation

$$\log P = -1.4 \log 10 + 2.5 = -1.4 + 2.5 = 1.1$$

Substitute $\log V = \log 10 = 1$ into the linearised equation. $\log 10$ is exactly 1 in base 10 — a useful shortcut.

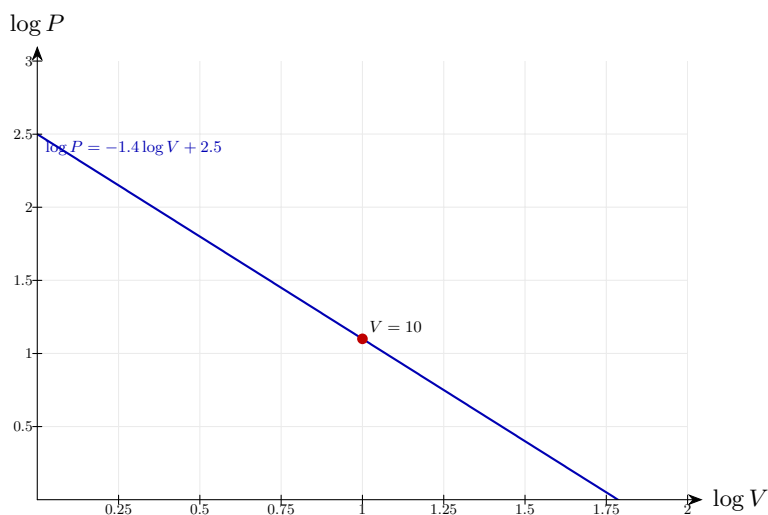
Step 2: Find P

$$P = 10^{1.1} = 12.6$$

Apply $10^{\square}$ to recover P from $\log P$.

Final answer: $P = 12.6$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 65 (e16-065)

Difficulty: Foundation / Marks: 4

A graph of $\log y$ against x has equation $\log y = -0.15x + 4$. Find the half-life (when $y =$ half initial value).

Worked solution.

Step 1: Initial value at $x=0$

$$\log y_0 = 4 \implies y_0 = 10000$$

The vertical intercept gives the initial value: at $x = 0$, $\log y = 4$, so $y_0 = 10^4$.

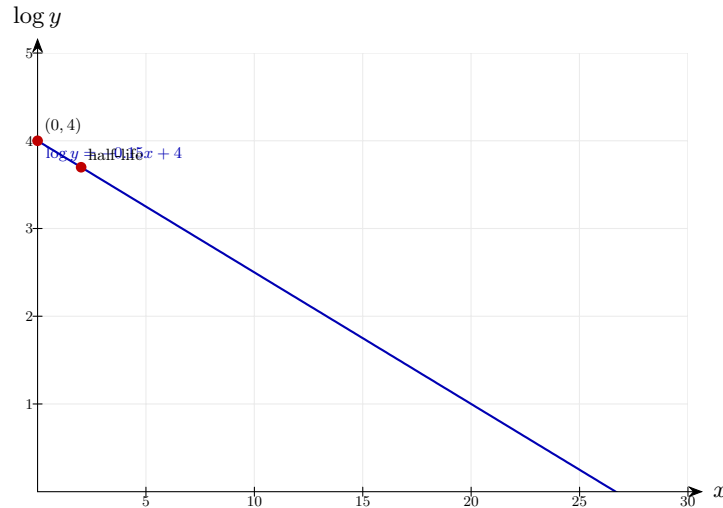
Step 2: Half value

$$\log 5000 = -0.15x + 4 \implies 3.699 = -0.15x + 4 \implies x = 2.01$$

Set $y = y_0/2 = 5000$, take $\log$ ($\log 5000 \approx 3.699$), and solve the linear equation. The negative gradient confirms decay.

Final answer: Half-life = 2.01 units

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 66 (e16-066)

Difficulty: Foundation / Marks: 3

Show that $y = 2x^3$ and $y = 8x$ intersect when $x = 2$ by using logarithmic methods.

Worked solution.

Step 1: Take log of both

$$\log 2 + 3 \log x = \log 8 + \log x$$

Set $2x^3 = 8x$ and take $\log$ of both sides. The product law splits each side into a constant plus a $\log$ term, and the power law brings the exponent down as a coefficient — turning a polynomial equation into a linear one in $\log x$.

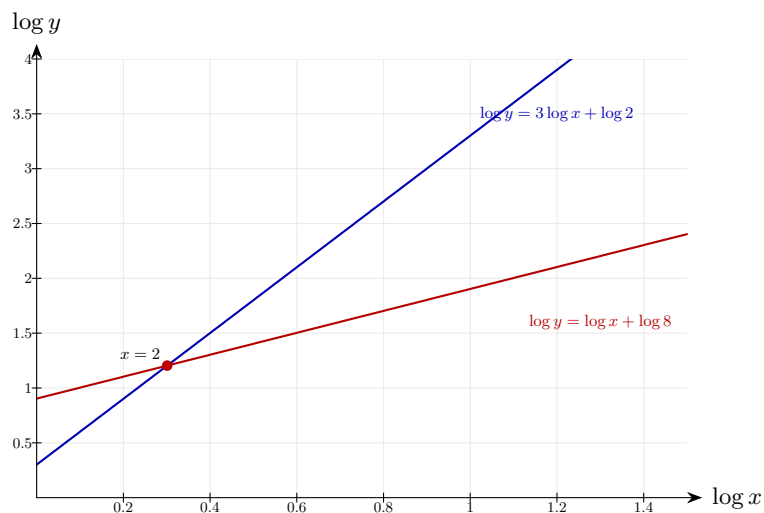
Step 2: Solve

$$2 \log x = \log 8 - \log 2 = \log 4 \implies \log x = \log 2 \implies x = 2$$

Collect the $\log x$ terms, use the quotient law to combine the constants ($\log 8 - \log 2 = \log 4$), and recognise $\log 4 = 2 \log 2$, giving $\log x = \log 2$, so $x = 2$.

Final answer: $x = 2$ (verified)

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 67 (e16-067)

Difficulty: Foundation / Marks: 5

The time t (s) for a pendulum to swing depends on length l (m): $t = al^n$. A plot of $\log t$ vs $\log l$ gives gradient 0.5 and intercept 0.3. Find a and n , and find t when $l = 4$.

Worked solution.

Step 1: Read off

$$n = 0.5, \quad a = 10^{0.3} = 2.0$$

Power model on $\log t$ vs $\log l$: gradient = n directly, intercept = $\log a$. Apply $10^{\square}$ to recover a . This matches the physical relationship $T \propto \sqrt{l}$ for a pendulum.

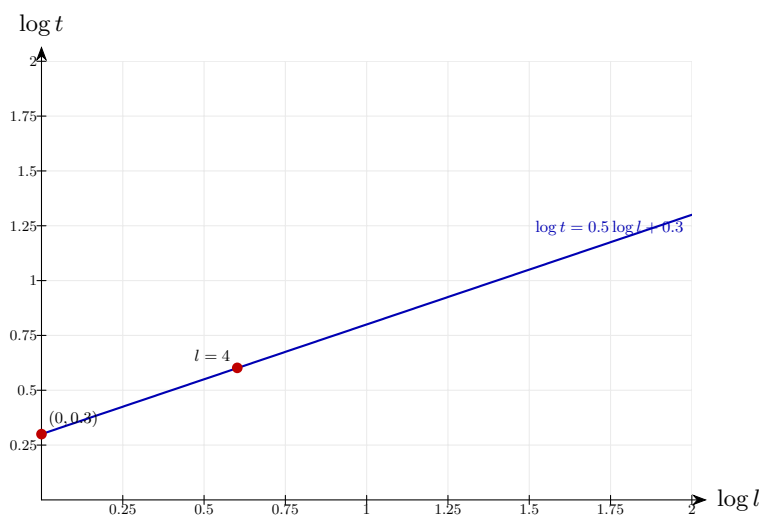
Step 2: Find t at $l=4$

$$t = 2\sqrt{4} = 4 \text{ s}$$

Substitute into $t = 2l^{0.5} = 2\sqrt{l}$. At $l = 4$, $\sqrt{4} = 2$, so $t = 4$ s.

Final answer: $a = 2, n = 0.5; t(4) = 4\text{s}$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 68 (e16-068)

Difficulty: Foundation / Marks: 5

A graph of $\ln N$ against t has equation $\ln N = 0.2t + 5$. Find: (a) the initial value; (b) the doubling time.

Worked solution.

Step 1: (a) $t = 0$

$$\ln N = 5 \implies N = e^5 = 148.4$$

Substitute $t = 0$. Apply $e^{\square}$ (not $10^{\square}$) because the graph uses $\ln$, not $\log_{10}$. Keeping the base consistent is the key trap to avoid.

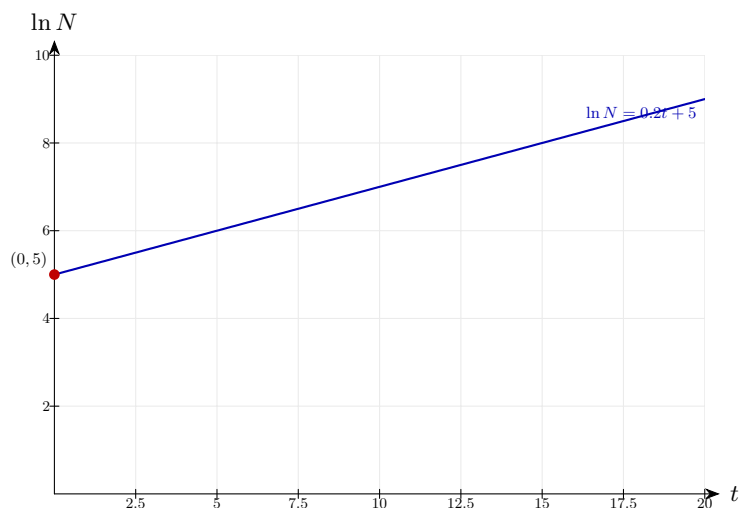
Step 2: (b) When N doubles

$$\ln(2N_0) = 0.2t + 5 \implies \ln 2 = 0.2(t - 0) \implies t = \frac{\ln 2}{0.2} = 3.47$$

Doubling means $\ln(2N_0) - \ln N_0 = \ln 2$; on the linearised graph this is a rise of $\ln 2$, so the time taken is $\ln 2/\text{gradient}$. For any exponential, doubling time is constant.

Final answer: (a) 148.4 (b) 3.47 time units

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 69 (e16-069)

Difficulty: Foundation / Marks: 2

Two models are proposed for data: Model A $y = ax^n$ and Model B $y = kb^x$. Plotting $\log y$ vs $\log x$ gives a curve but $\log y$ vs x gives a line. Which model fits?

Worked solution.

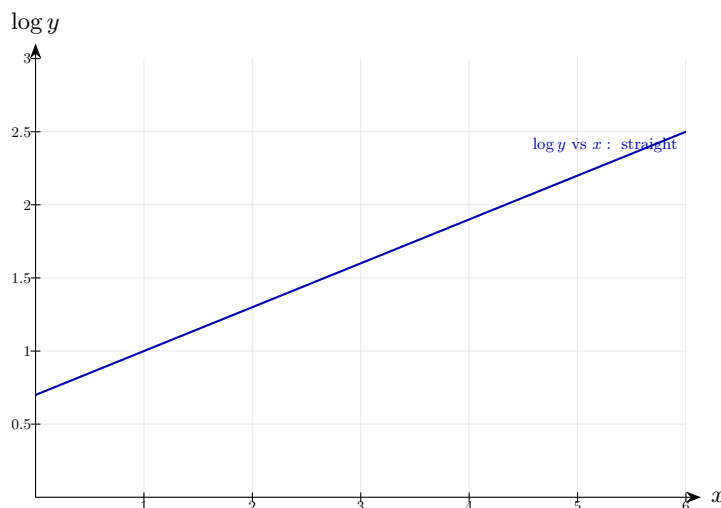
Analysis

Model B fits: $y = kb^x$

A power model gives a line on $\log y$ vs $\log x$; an exponential model gives a line on $\log y$ vs x . Here the second linearisation is the straight one, so the data is exponential.

Final answer: Model B: $y = kb^x$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 70 (e16-070)

Difficulty: Foundation / Marks: 6

A graph of $\log y$ against $\log x$ passes through $(0, 1.5)$ and $(2, 4.5)$. Find the equation $y = ax^n$, then find: (a) y when $x = 100$; (b) x when $y = 1000$.

Worked solution.

Step 1: Find n

$$n = \frac{4.5 - 1.5}{2 - 0} = 1.5$$

Gradient = $\Delta(\log y)/\Delta(\log x) = n$.

Step 2: Find a

$$\log a = 1.5 \implies a = 10^{1.5} \approx 31.6$$

The first point $(0, 1.5)$ lies on the $\log y$ -axis (where $\log x = 0$), so it gives $\log a$ directly.

Step 3: (a) $x = 100$

$$y = 31.6 \times 100^{1.5} = 31.6 \times 1000 = 31600$$

$100^{1.5} = (10^2)^{1.5} = 10^3 = 1000$. A clean integer power makes this easy without a calculator.

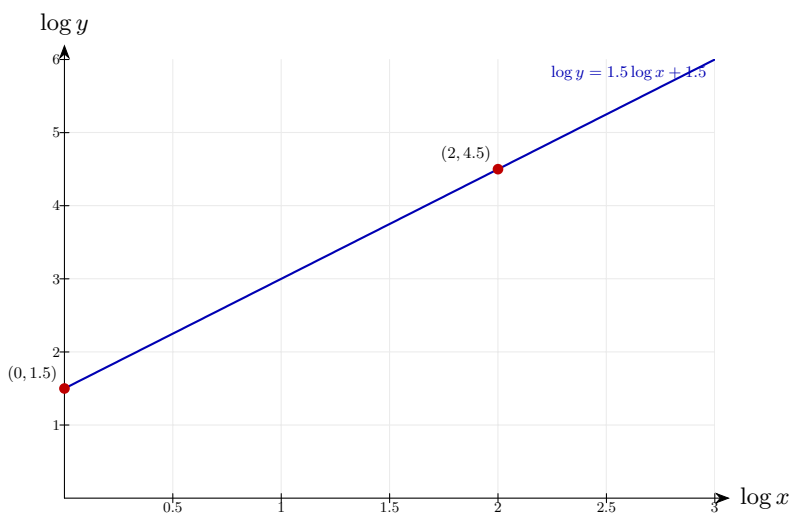
Step 4: (b) $y = 1000$

$$1000 = 31.6x^{1.5} \implies x^{1.5} = 31.6 \implies x = 31.6^{2/3} = 10.0$$

Isolate $x^{1.5}$, then raise to the power $2/3$ (the reciprocal of 1.5) to solve for x . Since $31.6 \approx 10^{1.5}$, we get $x \approx (10^{1.5})^{2/3} = 10$.

Final answer: $y = 31.6x^{1.5}$; (a) 31600 (b) $x = 10$

Diagram.



The linearised straight-line graph, with gradient, intercept and key points marked.

Using Logarithmic Graphs 71 (e16-071)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = e^x$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Identify the key features.

y -intercept: $e^0 = 1$ asymptote: $y = 0$ as $x \rightarrow -\infty$

Any exponential $y = a^x$ with $a > 1$ passes through $(0, 1)$ since $a^0 = 1$, and tends to 0 (but never reaches it) as $x \rightarrow -\infty$, giving a horizontal asymptote at $y = 0$.

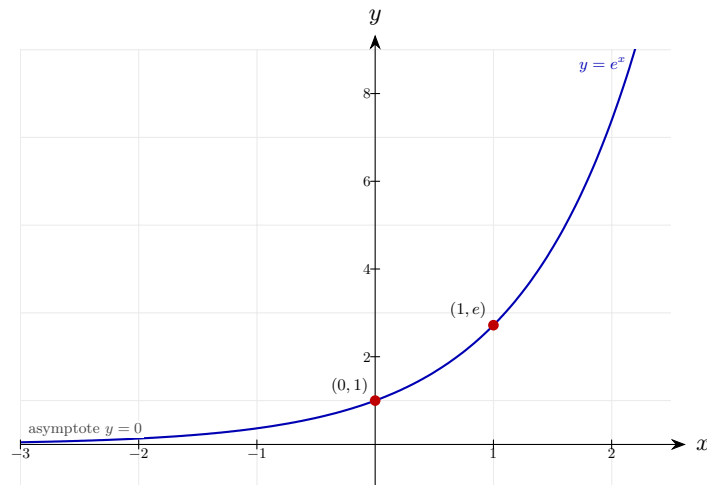
Step 2: Sketch the curve.

$y = e^x$ is increasing for all x ; $y > 0$ everywhere

The curve is always positive and increases without bound — steeply for $x > 0$, shallowly for $x < 0$. At $x = 1$ it reaches $e \approx 2.72$.

Final answer: Increasing curve through $(0, 1)$ and $(1, e)$, with horizontal asymptote $y = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 72 (e16-072)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = \ln x$, labelling the x -intercept and any asymptote.

Worked solution.

Step 1: Identify the key features.

$$x\text{-intercept: } \ln 1 = 0 \quad \text{asymptote: } x = 0 \text{ as } x \rightarrow 0^+$$

The natural log is the inverse of e^x : it is only defined for $x > 0$, passes through $(1, 0)$ since $\ln 1 = 0$, and goes to $-\infty$ as $x \rightarrow 0^+$, giving a vertical asymptote at $x = 0$.

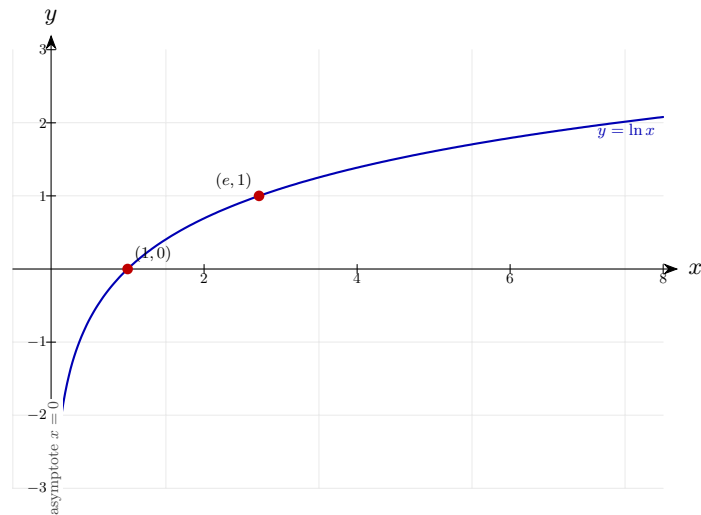
Step 2: Sketch the curve.

$$y = \ln x \text{ is increasing for all } x > 0; \quad \ln e = 1$$

Growth is slow once past $x = 1$ — for example $\ln e \approx 1$, $\ln 10 \approx 2.30$. The curve is the reflection of $y = e^x$ in the line $y = x$.

Final answer: Increasing curve through $(1, 0)$ and $(e, 1)$, with vertical asymptote $x = 0$; defined only for $x > 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 73 (e16-073)

Difficulty: Foundation / Marks: 4

On the same axes, sketch $y = e^x$ and $y = e^{-x}$. State the relationship between the two curves.

Worked solution.

Step 1: Identify the key features of each.

$$y = e^x : (0, 1), \text{ increasing} \quad y = e^{-x} : (0, 1), \text{ decreasing}$$

Both pass through $(0, 1)$ because $e^0 = 1$. e^x grows as x increases; e^{-x} decays as x increases (because the exponent gets more negative).

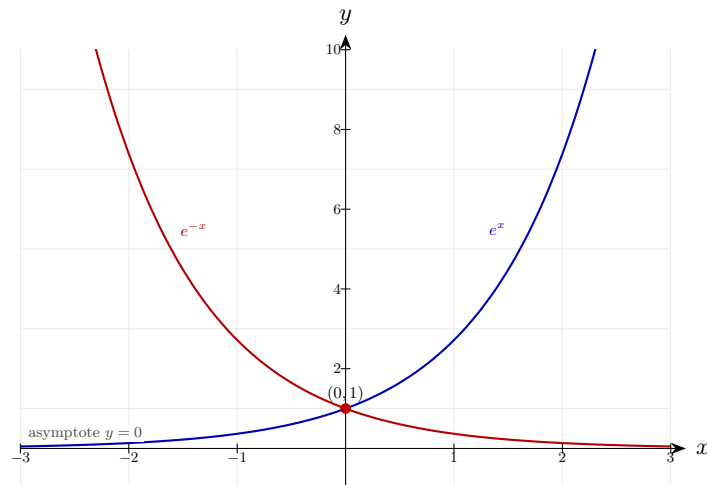
Step 2: Relate them by symmetry.

$$e^{-x} = e^{(-1) \cdot x}$$

Replacing x with $-x$ reflects the curve in the y -axis, so $y = e^{-x}$ is the mirror image of $y = e^x$ about the y -axis. Both have horizontal asymptote $y = 0$.

Final answer: Both curves pass through $(0, 1)$ with asymptote $y = 0$; $y = e^{-x}$ is the reflection of $y = e^x$ in the y -axis.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 74 (e16-074)

Difficulty: Foundation / Marks: 4

Sketch the curve $y = e^x - 1$, labelling the x - and y -intercepts and the asymptote.

Worked solution.

Step 1: Recognise the transformation.

$y = e^x - 1$ is a vertical translation of $y = e^x$ down by 1

Subtracting a constant from y shifts the whole curve down by that amount; the shape and steepness are unchanged, only the vertical position changes.

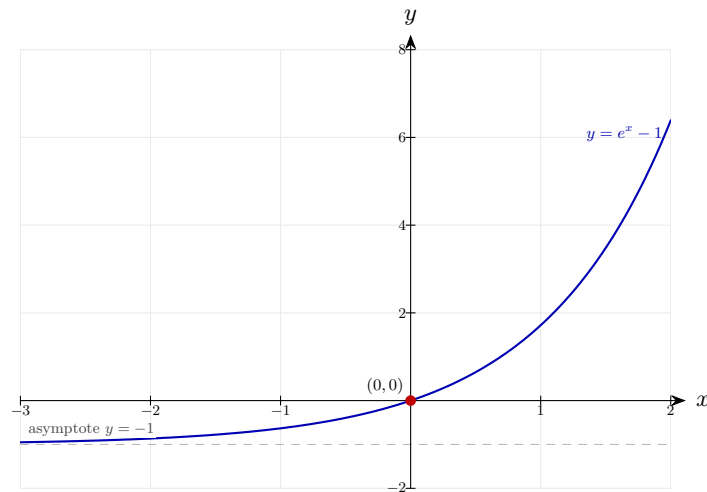
Step 2: Find the new intercepts and asymptote.

$$y\text{-int: } e^0 - 1 = 0 \quad x\text{-int: } e^x = 1 \Rightarrow x = 0 \quad \text{asymptote: } y = -1$$

The curve now passes through the origin since both intercepts coincide at $(0, 0)$. The horizontal asymptote shifts from $y = 0$ down to $y = -1$.

Final answer: Curve through the origin $(0, 0)$ with horizontal asymptote $y = -1$; the standard exponential shifted down by 1.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 75 (e16-075)

Difficulty: Standard / Marks: 4

Sketch the curve $y = \ln(x + 2)$, labelling the x -intercept, y -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$y = \ln(x + 2)$ is a horizontal translation of $y = \ln x$ to the left by 2

Replacing x with $x + 2$ shifts the curve in the $-x$ direction by 2. The vertical asymptote moves with it from $x = 0$ to $x = -2$.

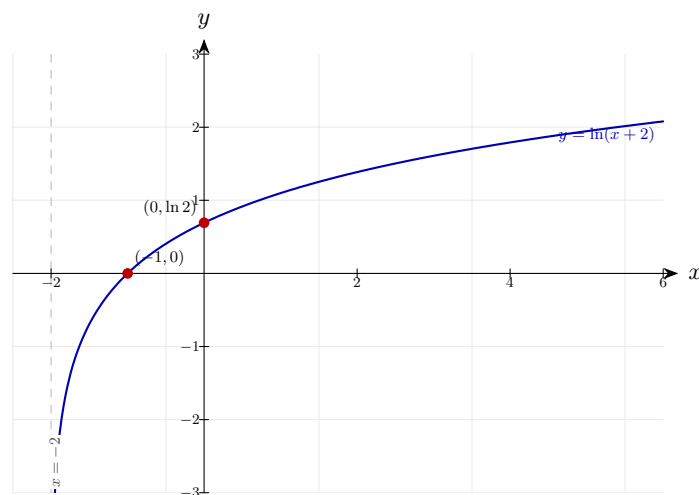
Step 2: Find the intercepts.

x -int: $\ln(x + 2) = 0 \Rightarrow x + 2 = 1 \Rightarrow x = -1$ y -int: $\ln 2 \approx 0.69$

For the x -intercept, set $y = 0$ and use $\ln 1 = 0$. For the y -intercept, evaluate at $x = 0$, giving $\ln 2$.

Final answer: Curve through $(-1, 0)$ and $(0, \ln 2)$, with vertical asymptote $x = -2$; the standard $\ln$ shifted left by 2.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 76 (e16-076)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = 2^x$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Identify the key features.

$$y\text{-int: } 2^0 = 1 \quad \text{asymptote: } y = 0 \text{ as } x \rightarrow -\infty$$

Any exponential $y = a^x$ with $a > 1$ has the same general shape: passes through $(0, 1)$ and tends to 0 as $x \rightarrow -\infty$. The base $2 < e$, so 2^x grows more slowly than e^x .

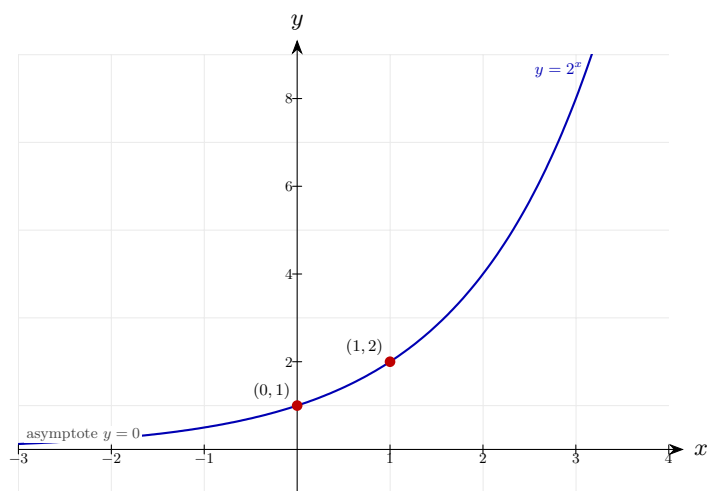
Step 2: Sketch the curve.

$$2^1 = 2, \quad 2^2 = 4, \quad 2^{-1} = \frac{1}{2}$$

Plot a few values and join with a smooth increasing curve. Always positive, never crosses the x -axis.

Final answer: Increasing curve through $(0, 1)$ and $(1, 2)$, with horizontal asymptote $y = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 77 (e16-077)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = \log_{10} x$, labelling the x -intercept and any asymptote.

Worked solution.

Step 1: Identify the key features.

$$x\text{-int: } \log_{10} 1 = 0 \quad \text{asymptote: } x = 0 \quad \log_{10} 10 = 1$$

Any $\log y = \log_a x$ with $a > 1$ passes through $(1, 0)$ and has vertical asymptote $x = 0$. Growth is slower than $\ln x$ since $\log_{10} = \ln / \ln 10$ and $\ln 10 \approx 2.30$.

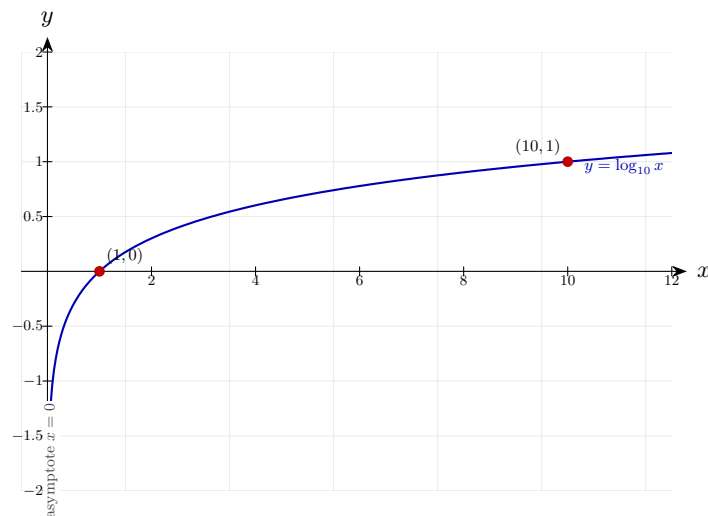
Step 2: Sketch the curve.

$$\log_{10} 10 = 1, \quad \log_{10} 100 = 2, \quad \log_{10} 0.1 = -1$$

Defined only for $x > 0$. Note how slowly the curve climbs once past $x = 1$ — it takes a tenfold increase in x to add 1 to y .

Final answer: Increasing curve through $(1, 0)$ and $(10, 1)$, with vertical asymptote $x = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 78 (e16-078)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = -e^x$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = -e^x \text{ is a reflection of } y = e^x \text{ in the } x\text{-axis}$$

Multiplying y by -1 flips the curve about the x -axis: every positive output becomes a negative output of the same magnitude.

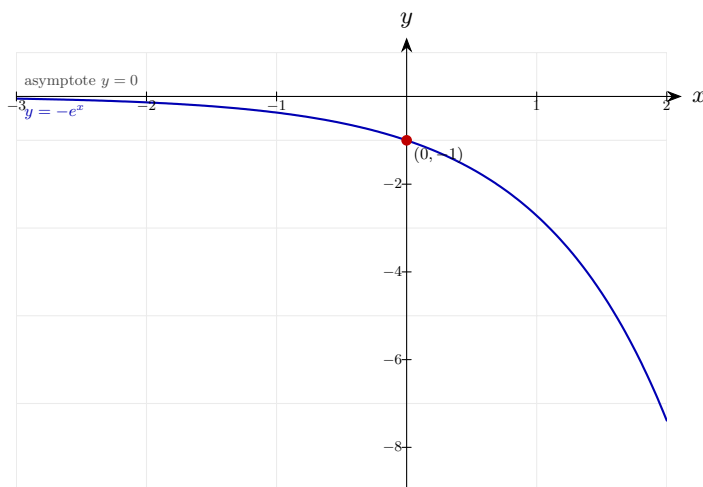
Step 2: Find intercepts and asymptote.

$$y\text{-int: } -e^0 = -1 \quad \text{asymptote: } y = 0 \text{ as } x \rightarrow -\infty$$

Now y is always negative, decreasing without bound as x increases. The asymptote $y = 0$ is approached from below.

Final answer: Decreasing curve through $(0, -1)$, always negative, with horizontal asymptote $y = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 79 (e16-079)

Difficulty: Standard / Marks: 4

Sketch the curve $y = e^{2x}$, labelling the y -intercept and any asymptote. Compare its steepness with $y = e^x$ at $x = 1$.

Worked solution.

Step 1: Recognise the transformation.

$y = e^{2x}$ is a horizontal compression of $y = e^x$ by factor $\frac{1}{2}$

Replacing x with $2x$ compresses the curve horizontally by $\frac{1}{2}$, so growth happens twice as fast: e^{2x} at $x = 1$ equals e^x at $x = 2$.

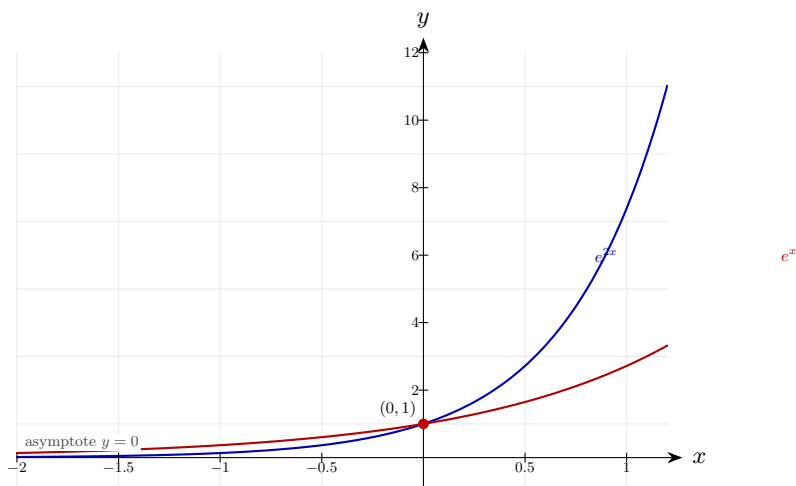
Step 2: Find intercepts and asymptote.

y -int: $e^0 = 1$ asymptote: $y = 0$ $e^{2(1)} = e^2 \approx 7.39$

The y -intercept and asymptote are unchanged from $y = e^x$; only the rate of growth changes. At $x = 1$, this curve has already reached e^2 , where e^x only reaches $e \approx 2.72$.

Final answer: Increasing curve through $(0, 1)$ and $(1, e^2)$, with horizontal asymptote $y = 0$; grows twice as fast as $y = e^x$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 80 (e16-080)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = e^x + 2$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = e^x + 2 \text{ is } y = e^x \text{ translated up by } 2$$

Adding a constant to y shifts the whole curve vertically; the shape and steepness are unchanged.

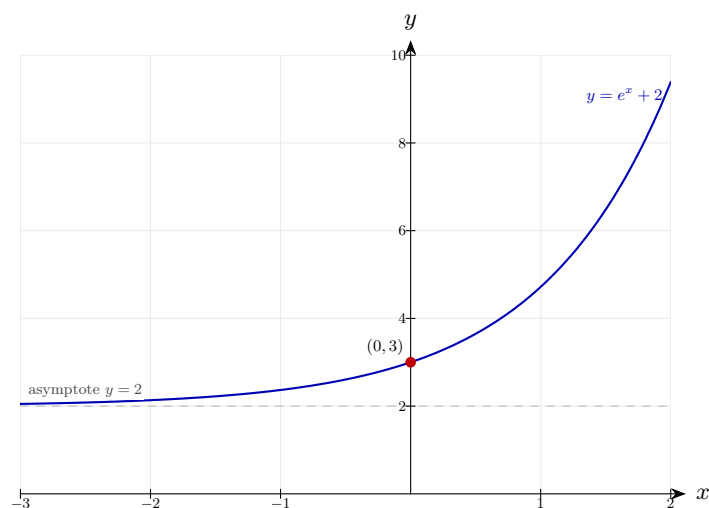
Step 2: Find intercepts and asymptote.

$$y\text{-int: } e^0 + 2 = 3 \quad \text{asymptote: } y = 2 \text{ as } x \rightarrow -\infty$$

There is no x -intercept since $e^x > 0$ for all x , so $y \geq 2$ always. The horizontal asymptote shifts up from $y = 0$ to $y = 2$.

Final answer: Increasing curve through $(0, 3)$ with horizontal asymptote $y = 2$; no x -intercept.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 81 (e16-081)

Difficulty: Standard / Marks: 4

Sketch the curve $y = e^{-x} + 1$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = e^{-x} + 1 \text{ is } y = e^{-x} \text{ translated up by } 1$$

Start from the standard decay curve $y = e^{-x}$ (reflection of e^x in the y -axis), then shift everything up by 1.

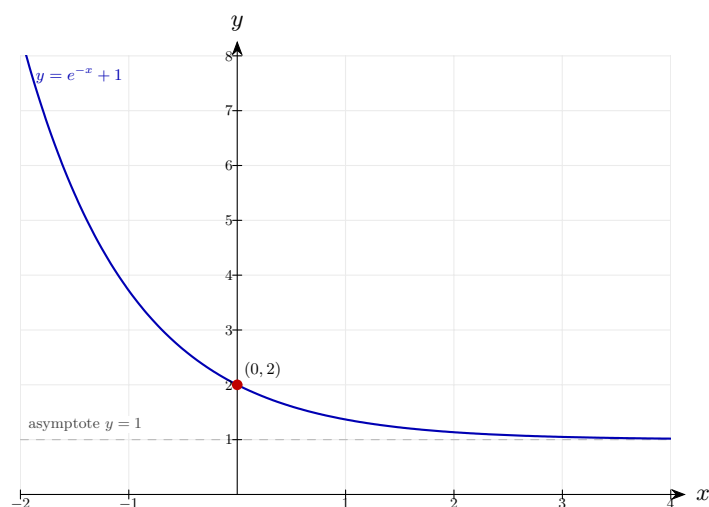
Step 2: Find intercepts and asymptote.

$$y\text{-int: } e^0 + 1 = 2 \quad \text{asymptote: } y = 1 \text{ as } x \rightarrow \infty$$

The curve decreases from large positive values on the left towards $y = 1$ on the right. There is no x -intercept since $e^{-x} > 0$ gives $y > 1$ always.

Final answer: Decreasing curve through $(0, 2)$ with horizontal asymptote $y = 1$; no x -intercept.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 82 (e16-082)

Difficulty: Standard / Marks: 4

Sketch the curve $y = \ln(2x)$, labelling the x -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$\ln(2x) = \ln 2 + \ln x$$

Using the product law of logs, $\ln(2x)$ is just $\ln x$ shifted vertically up by the constant $\ln 2 \approx 0.693$. The shape and asymptote of $\ln x$ are unchanged.

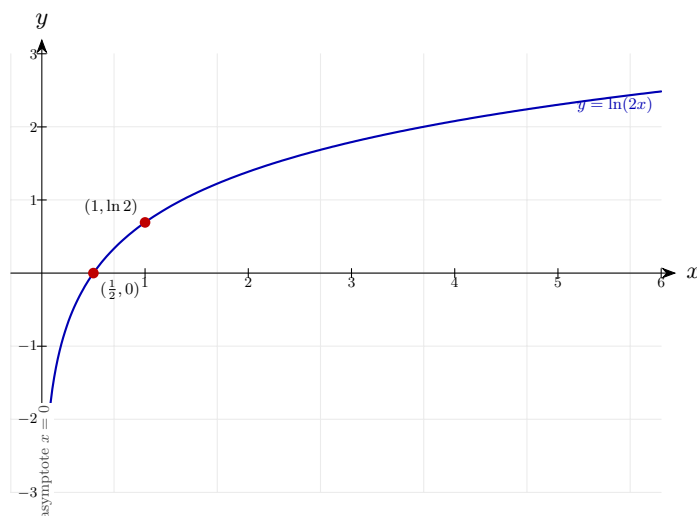
Step 2: Find intercept and asymptote.

$$x\text{-int: } \ln(2x) = 0 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2} \quad \text{asymptote: } x = 0$$

The vertical asymptote stays at $x = 0$ because the transformation acts on y , not on x . The x -intercept moves to $x = \frac{1}{2}$ since $2x = 1$ there.

Final answer: Increasing curve through $(\frac{1}{2}, 0)$ and $(1, \ln 2)$, with vertical asymptote $x = 0$; equivalent to $\ln x$ shifted up by $\ln 2$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 83 (e16-083)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = \ln x + 1$, labelling the x -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = \ln x + 1 \text{ is } y = \ln x \text{ translated up by } 1$$

Vertical translation by $+1$; shape and vertical asymptote at $x = 0$ are unchanged.

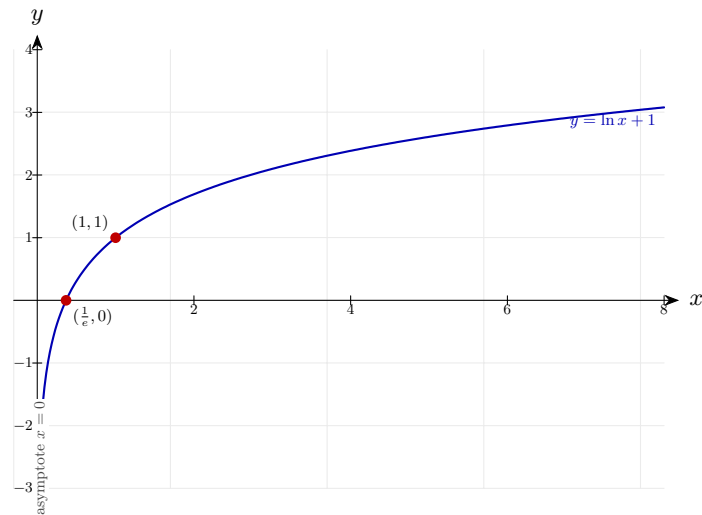
Step 2: Find intercept and asymptote.

$$x\text{-int: } \ln x + 1 = 0 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1} = \frac{1}{e} \quad \text{asymptote: } x = 0$$

Solve $y = 0$ by taking $e^{\square}$ of both sides; this gives $x = 1/e \approx 0.368$. The x -intercept shifts left from 1 since the curve has moved up.

Final answer: Increasing curve through $(\frac{1}{e}, 0)$ and $(1, 1)$, with vertical asymptote $x = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 84 (e16-084)

Difficulty: Standard / Marks: 4

Sketch the curve $y = \ln(-x)$, stating its domain and labelling any asymptote and intercept.

Worked solution.

Step 1: Determine the domain.

$$-x > 0 \implies x < 0$$

A logarithm requires a positive argument, so $-x > 0$ forces $x < 0$. The curve only exists to the left of the y -axis.

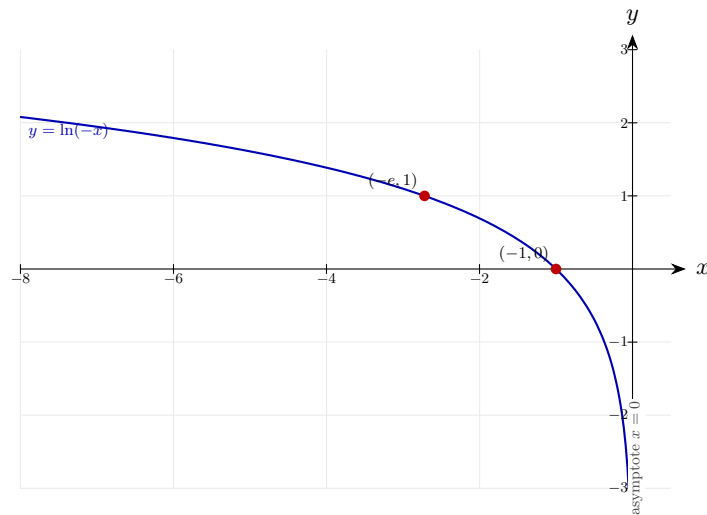
Step 2: Recognise the transformation and find features.

$$y = \ln(-x) \text{ is a reflection of } y = \ln x \text{ in the } y\text{-axis}$$

Replacing x with $-x$ reflects the standard $\ln x$ curve in the y -axis. The asymptote stays at $x = 0$; the x -intercept moves from $(1, 0)$ to $(-1, 0)$ since $\ln(-(-1)) = \ln 1 = 0$.

Final answer: Defined only for $x < 0$; curve through $(-1, 0)$ and $(-e, 1)$, with vertical asymptote $x = 0$; reflection of $\ln x$ in the y -axis.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 85 (e16-085)

Difficulty: Standard / Marks: 5

Sketch the curve $y = 3 - e^x$, labelling the x - and y -intercepts and the asymptote.

Worked solution.

Step 1: Recognise the transformations.

$$y = 3 - e^x = -e^x + 3$$

Two transformations of $y = e^x$: reflect in the x -axis (the $-e^x$ part), then translate up by 3. The asymptote at $y = 0$ reflects to itself, then shifts up to $y = 3$.

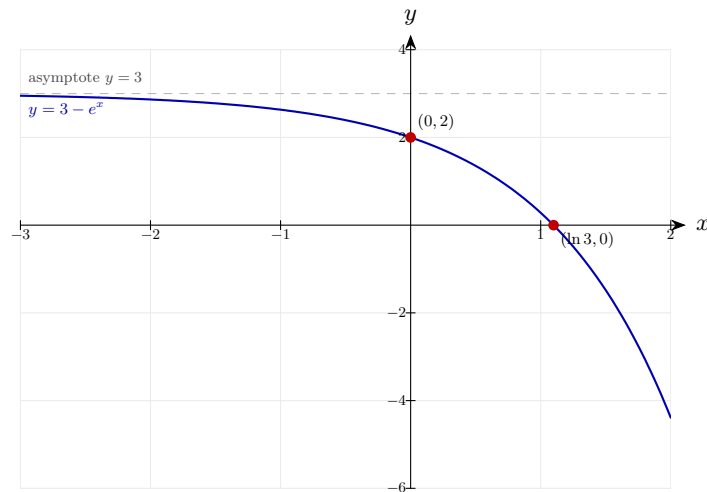
Step 2: Find intercepts and asymptote.

$$y\text{-int: } 3 - e^0 = 2 \quad x\text{-int: } e^x = 3 \Rightarrow x = \ln 3 \approx 1.10 \quad \text{asymptote: } y = 3$$

For the x -intercept, solve $y = 0$ and take $\ln$ of both sides. The curve approaches $y = 3$ from below as $x \rightarrow -\infty$ and decreases without bound as $x \rightarrow \infty$.

Final answer: Decreasing curve through $(0, 2)$ and $(\ln 3, 0)$, with horizontal asymptote $y = 3$; a reflection-and-shift of $y = e^x$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 86 (e16-086)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = e^{x-1}$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = e^{x-1} \text{ is } y = e^x \text{ translated right by 1}$$

Replacing x with $x-1$ shifts the curve in the $+x$ direction by 1. The asymptote at $y = 0$ is unchanged because it is horizontal.

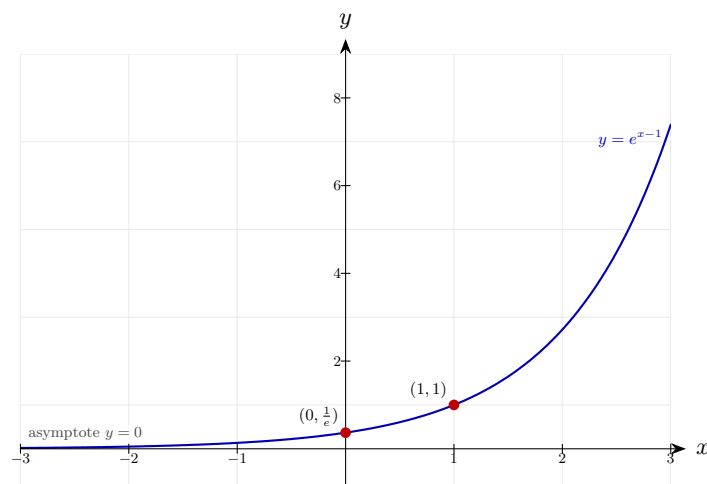
Step 2: Find the intercepts.

$$y\text{-int: } e^{0-1} = e^{-1} = \frac{1}{e} \approx 0.368 \quad \text{asymptote: } y = 0$$

No x -intercept since $e^{x-1} > 0$ always. The curve still passes through the point one unit right of the original $(0, 1)$: namely $(1, 1)$.

Final answer: Increasing curve through $(0, \frac{1}{e})$ and $(1, 1)$, with horizontal asymptote $y = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 87 (e16-087)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = \ln(x - 1)$, labelling the x -intercept and any asymptote.

Worked solution.

Step 1: Recognise the transformation.

$$y = \ln(x - 1) \text{ is } y = \ln x \text{ translated right by } 1$$

Replacing x with $x - 1$ shifts the whole curve in the $+x$ direction by 1. The vertical asymptote shifts with it from $x = 0$ to $x = 1$.

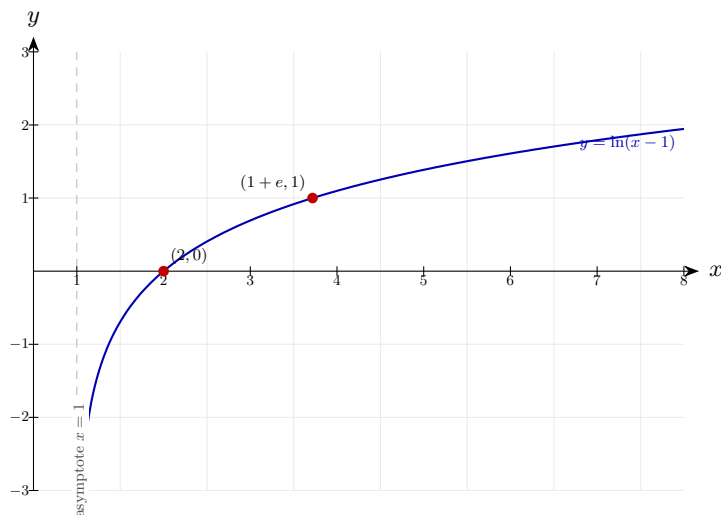
Step 2: Find intercept and asymptote.

$$x\text{-int: } \ln(x - 1) = 0 \Rightarrow x - 1 = 1 \Rightarrow x = 2 \quad \text{asymptote: } x = 1 \quad \text{domain: } x > 1$$

Defined only for $x > 1$ because the log requires a positive argument. The intercept moves from $(1, 0)$ to $(2, 0)$.

Final answer: Increasing curve through $(2, 0)$ and $(1 + e, 1)$, with vertical asymptote $x = 1$; defined only for $x > 1$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 88 (e16-088)

Difficulty: Standard / Marks: 4

Sketch the curve $y = 2 \ln x$, labelling the x -intercept and any asymptote. Compare with $y = \ln x$ at $x = e$.

Worked solution.

Step 1: Recognise the transformation.

$y = 2 \ln x$ is a vertical stretch of $y = \ln x$ by factor 2

Multiplying y by 2 stretches every output to twice its value; the shape is the same but the curve climbs faster. The vertical asymptote at $x = 0$ is unaffected by a vertical stretch.

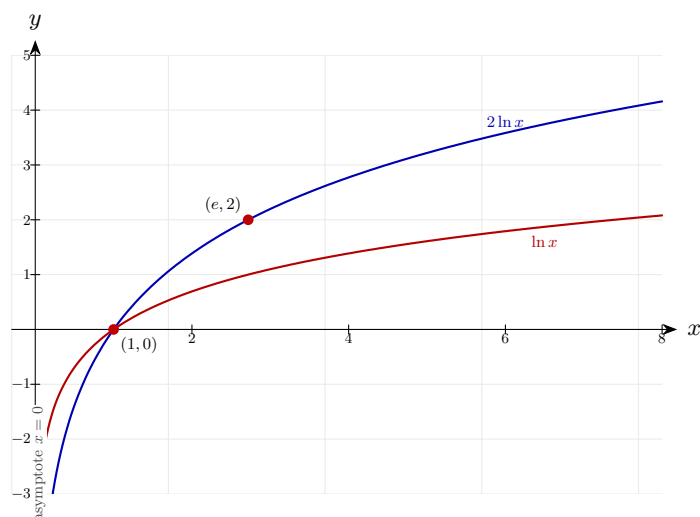
Step 2: Find intercept and asymptote.

$$x\text{-int: } 2 \ln x = 0 \Rightarrow \ln x = 0 \Rightarrow x = 1 \quad \text{asymptote: } x = 0 \quad 2 \ln e = 2$$

The x -intercept is unchanged at $(1, 0)$ — multiplying 0 by anything is still 0. At $x = e$ this curve reaches 2, where $\ln x$ only reaches 1.

Final answer: Increasing curve through $(1, 0)$ and $(e, 2)$, with vertical asymptote $x = 0$; twice as tall as $y = \ln x$ for every $x > 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 89 (e16-089)

Difficulty: Foundation / Marks: 3

Sketch the curve $y = 2^{-x}$, labelling the y -intercept and any asymptote.

Worked solution.

Step 1: Rewrite as a decay.

$$2^{-x} = \left(\frac{1}{2}\right)^x$$

A negative exponent inverts the base, so 2^{-x} is the same as $(1/2)^x$ — a decaying exponential with base less than 1.

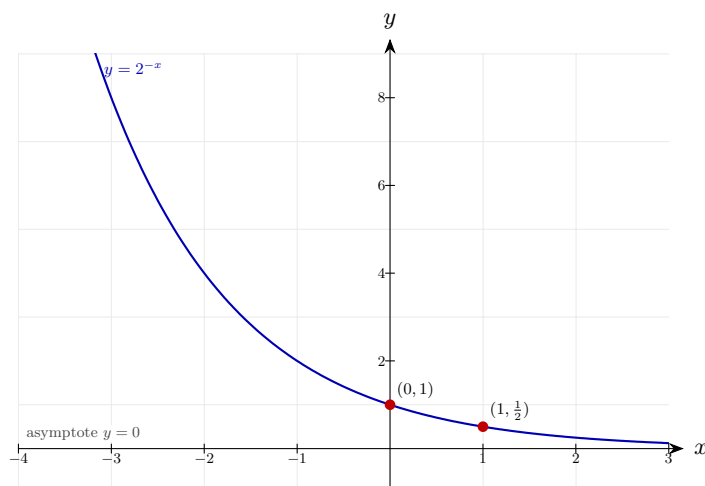
Step 2: Find intercept and asymptote.

$$y\text{-int: } 2^0 = 1 \quad \text{asymptote: } y = 0 \text{ as } x \rightarrow \infty$$

Passes through $(0, 1)$ and decays towards the x -axis as x increases. At $x = 1$ it reaches $\frac{1}{2}$; at $x = -1$ it climbs to 2.

Final answer: Decreasing curve through $(0, 1)$ and $(1, \frac{1}{2})$, with horizontal asymptote $y = 0$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.

Using Logarithmic Graphs 90 (e16-090)

Difficulty: Standard / Marks: 4

On the same axes, sketch $y = \ln x$ and $y = \log_{10} x$. Which curve climbs faster and why?

Worked solution.

Step 1: Identify shared and distinct features.

Both pass through $(1, 0)$; both have asymptote $x = 0$

Every $y = \log_a x$ with $a > 1$ has the same x -intercept $(1, 0)$ and the same vertical asymptote $x = 0$; only the rate of growth differs.

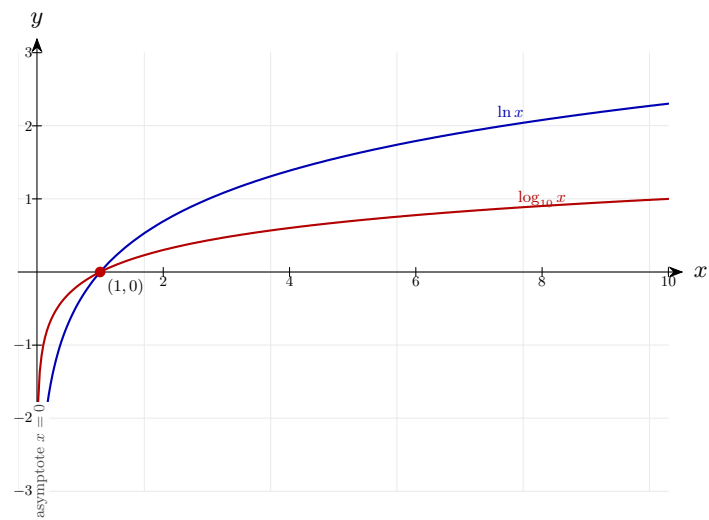
Step 2: Compare growth rates using change-of-base.

$$\log_{10} x = \frac{\ln x}{\ln 10} \approx \frac{\ln x}{2.30}$$

$\log_{10} x$ is just $\ln x$ scaled vertically by $1/\ln 10 \approx 0.434$, so it climbs slower. At $x = e$, $\ln x = 1$ but $\log_{10} x \approx 0.434$.

Final answer: Both pass through $(1, 0)$ with asymptote $x = 0$; $y = \ln x$ climbs faster because $\log_{10} x = \frac{\ln x}{\ln 10}$ and $\ln 10 \approx 2.30 > 1$.

Diagram.



The curve sketched with its intercepts and asymptote labelled.
